FROM LITERARY WARRANT TO COGNITIVE WARRANT

Extending Hybrid Meaning Theory to Professionally Imposed Categorical Boundaries

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This document is a psycholinguistics-focused version of a larger research program developed as my Master of Library and Information Science capstone project at the University of Denver (begun January 2026). The full research manuscript develops the complete theoretical framework, knowledge organization system (KOS) reform applications, and bibliometric extensions of the cognitive warrant construct. This version is a deliberate editorial selection from that larger architecture, retaining the experimental design, statistical analysis plan, and theoretical contributions most relevant to a psycholinguistics audience and condensing material whose primary audience is library and information science.

Three components of the full manuscript are condensed or deferred. The warrant-theory genealogy—distinguishing literary, user, and cultural warrant in the KOS evaluation tradition—is reduced to the minimum needed to motivate the cognitive warrant construct. The surrogate-assisted pCI-gradient ascent procedure, which extends per-item behavioral diagnostics into a prescriptive optimization algorithm for controlled vocabulary reform, is summarized in a single paragraph; its technical development is available in a companion document. The bibliometric and hyperbolic geometry extensions of the framework are not included.

What this version retains in full: the information-theoretic formalization of cognitive warrant as $\mathrm{CI}(d) \equiv I(Y; S \mid d)$; the connection to Trott and Bergen's hybrid meaning framework and the structural account of their ambiguity-type null result; the Library of Congress Subject Headings (LCSH) mapping methodology and librarian validation protocol; the NetworkX directed acyclic graph implementation of LCSH's polyhierarchy; the two-experiment behavioral design with complete `lme4` specifications; and the hypothesis set with its analysis plan.

Status of study: As of mid-June, IRB application #2446127-1 for exempt certification was still in the review process. This precluded, prior to finishing the capstone course, (1) expert consultant validation of LCSH stimulus mapping by two librarians, and (2) pilot testing of Experiments 1 and 2 with SONA participants via the Pavlovia platform. A proof of concept to demonstrate the plausibility of cognitive warrant has been conducted, however, and the Preliminary Results chapter provides the results of that analysis.

Grey boxes throughout the document contain pertinent highlights of material that has been abridged. All abridged material is available on request.

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Title: COGNITIVE WARRANT: EXTENDING HYBRID MEANING THEORY TO PROFESSIONALLY IMPOSED CATEGORICAL BOUNDARIES

ABSTRACT

Knowledge organization systems such as the Library of Congress Subject Headings (LCSH) structure the conceptual landscape through which users encounter information, imposing categorical sense distinctions that transform graded patterns of language use into stable, professionally authorized taxonomic structures. Their evaluation has traditionally proceeded through literary warrant, terminological consistency, and coverage—criteria derived from the published literature and cataloging tradition rather than from empirical evidence about human semantic processing. This research program introduces *cognitive warrant* as a complementary evaluation principle, defined as the extent to which the categorical and hierarchical distinctions encoded in a knowledge organization system correspond to observable patterns in human semantic cognition.

The framework is grounded in Trott and Bergen's (2023) hybrid model of lexical meaning—the finding that word meaning is simultaneously continuously graded (with processing difficulty proportional to semantic distance) and discretely categorical (producing processing discontinuities at sense boundaries). The central theoretical contribution is the extension of this hybrid architecture to professionally imposed rather than lexicographically derived sense boundaries, treating LC subject heading assignments as substitutes for dictionary headword entries in the categorical operationalization. Cognitive warrant is formalized as conditional mutual information:

$\mathrm{CI}(d) \equiv I(Y; S \mid d)$, quantifying the reduction in uncertainty about sense identity that a categorical boundary contributes above and beyond continuous semantic distance. This specification yields a graduated, ratio-scale evaluation metric with a principled zero and a principled maximum derived from information-theoretic first principles, directly addressing the graduation problem Trott and Parkinson-Coombs (2026) identify in the construction of behavioral benchmarks.

A stratified sample of 25 ambiguous words (50 senses, 299 stimulus pairs) drawn from Trott and Bergen's validated psycholinguistic stimulus pool was mapped to LC subject headings, and hierarchical relationships among the resulting headings were formalized as a directed acyclic graph constructed from LC Linked Data Service records. A proof-of-concept secondary analysis of Trott and Bergen's Experiment 2 behavioral data demonstrates that the analytic pipeline is end-to-end executable on real linguistic material. The hierarchical connectivity indicator (h^{finite}) predicts ground-truth sense identity above and beyond BERT cosine distance ($\hat{\beta} = 5.303$, LRT $\chi^2(2) = 164.75$, $p < .001$), resolving an estimated $\overline{\mathrm{CI}}_h = 0.190$ bits of residual sense-identity uncertainty per stimulus pair on average—a warrant efficiency of $\bar{\eta}_h = 0.266$, just over a quarter of the disambiguation load left unresolved by continuous distance alone—providing preliminary evidence that LCSH's DAG structure encodes cognitively meaningful distinctions. The heading congruence indicator (`same_heading`) exhibits complete separation in this secondary dataset—a consequence of its definitional equivalence to the lexicographic outcome variable—motivating the behavioral experiments required to test cognitive warrant as a processing phenomenon rather than a structural one. Librarian validation of the LCSH stimulus mapping and behavioral data collection are pending IRB approval.

The research program makes four contributions: a formal definition of cognitive warrant as an empirically testable evaluation principle of controlled vocabularies alongside literary, user, and cultural warrant; a reusable methodology for mapping psycholinguistic stimuli to LCSH and extracting structural graph measures from its linked data representation; an information-theoretic formalization of cognitive warrant as uncertainty reduction with a mathematically derived evaluation metric; and a complete experimental design, hypothesis set, and statistical analysis plan for a fully powered confirmatory study. Together, these establish the foundation for an empirical research program treating controlled vocabularies as falsifiable models of conceptual structure.

Keywords— knowledge organization systems, Library of Congress Subject Headings, cognitive warrant, psycholinguistics, distributional semantics, lexical ambiguity, BERT embeddings, information theory, directed acyclic graph

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GLOSSARY OF KEY TERMS

Cognitive information. The reduction in uncertainty about sense-category membership achieved by incorporating categorical structure into a semantic representation, formally defined as $\mathrm{CI}(d) = H(Y \mid d) - H(Y \mid d, S)$, where $H(Y \mid d)$ is the conditional entropy of sense identity given semantic distance d and $H(Y \mid d, S)$ is the conditional entropy after introducing categorical status variable S. Cognitive information will be shown to be greatest at intermediate semantic distances where $H(Y \mid d)$ is maximized—the "ambiguous zone."

Cognitive warrant. The principle that the categorical distinctions encoded in a knowledge organization system (KOS) should correspond to empirically observable patterns in human semantic cognition. More formally, a conceptual boundary in a KOS possesses cognitive warrant if the distinction it encodes corresponds to a measurable discontinuity in human judgments of semantic similarity or meaning, above and beyond what is predicted by continuous semantic distance alone. Operationalized in this study as a significant positive coefficient on `same_heading` (Experiment 1) or a significant negative coefficient on `same_heading` (Experiment 2) after controlling for `distance_bert`.

Directed acyclic graph (DAG). A directed graph containing no sequence of edges that, followed in their indicated directions, returns to the starting node. The graph representation of LCSH in this study is a DAG: directed because broader/narrower term relationships are asymmetric, and acyclic under the assumption that genuine hierarchical cycles are cataloging errors. The DAG representation supports the Lowest Common Ancestor algorithm and shortest-path computations required for `hier_distance`.

Discovery system. Integrated library search platforms (e.g., Ex Libris Primo) that provide unified keyword searching across multiple databases while incorporating controlled vocabulary through relevance boosting and faceted filtering. Modern discovery systems blend keyword and subject search approaches through hybrid ranking algorithms, meaning LCSH categorical boundaries influence retrieval even when users do not explicitly use subject search.

Discrimination score (Δr). The primary dependent variable in Experiment 2, defined as the appropriateness rating for the target LCSH heading minus the appropriateness rating for the within-category foil heading: $\Delta r = r_{\text{target}} - r_{\text{within-cat foil}}$. A positive Δr indicates that participants prefer the professionally assigned heading over a structurally defined foil at matched BERT distance—the operationalization of cataloger–user categorical alignment.

Entropy. A measure of the uncertainty about the value of a random variable or how surprised you should expect to be by the outcome of a random process. The more unlikely an outcome, the more information you gain when you observe it. Entropy, like many other concepts in my theoretical framework, comes from information theory (Shannon, 2001).

Experiment 1 replicates Trott and Bergen's (2023) primed sensibility judgment paradigm, substituting LCSH boundaries for dictionary-derived boundaries.

Experiment 2 introduces a heading appropriateness rating paradigm. After reading two sentences simultaneously establishing a specific word sense, participants rate (1–7 scale) how appropriate different LC subject headings would be for indexing documents on that topic. The dependent variable is the discrimination score Δr defined as the appropriateness rating for the target LC subject heading minus the rating for the within-neighborhood foil heading. A positive Δr indicates that participants prefer the professionally assigned heading over a structurally defined foil—the operationalization of cataloger–user categorical alignment.

Hierarchical distance. A quantitative measure of the proximity of two subject headings within the LCSH polyhierarchical graph, operationalized as the sum of shortest path lengths from each heading to their Lowest Common Ancestor (LCA) in the broader/narrower term subgraph. Hierarchical distance of 0 indicates the same heading; 1 indicates a direct broader/narrower term relationship; 2 indicates headings sharing an immediate broader term ("siblings"); ∞ indicates no common ancestor recoverable within the graph (typical for homonymous pairs crossing taxonomically distant LCSH subtrees).

LCSH (Library of Congress Subject Headings). The controlled vocabulary developed by the Library of Congress for subject cataloging in bibliographic records. Provides pre-coordinated subject strings combining topical, geographic, chronological, and form subdivisions (e.g., "Libraries–Academic–United States–History–20th century"). Authority records include explicit broader term (BT), narrower term (NT), related term (RT), and USE/UF cross-reference relationships. The BT/NT subgraph of LCSH constitutes the polyhierarchical directed acyclic graph (DAG) used in this study.

Literary warrant. The principle that controlled vocabulary structures should reflect what exists in published literature—new headings are added when sufficient publications on a topic accumulate. Literary warrant governs *when* to create a heading; cognitive warrant asks whether the categorical *placement* of that heading is cognitively defensible.

Lowest Common Ancestor (LCA). In the LCSH DAG, the most specific (deepest) subject heading that appears in the upward traversal paths of both target headings toward the root. Used to compute `hier_distance` via NetworkX's `lowest_common_ancestor` function operating on the BT/NT subgraph.

Parenthetical qualifier. Disambiguating text in parentheses appended to LCSH headings to distinguish polysemous or homonymous terms (e.g., *Java (Indonesia)* vs. *Java (Computer program language)*), or to clarify the intended referent.

Polyhierarchy. A property of LCSH whereby a single authority record may carry more than one Broader Term entry, allowing a concept to be classified under multiple parent nodes simultaneously. This structural property necessitates the DAG representation (rather than a tree) and means the LCA algorithm must search all available ancestral paths.

Scope note. Explanatory text in an LCSH authority record clarifying the intended usage of a heading, distinguishing it from related terms, and specifying what types of material should be assigned the heading. Scope notes are consulted during the heading selection phase of the mapping procedure. Note that only 2.5% of LCSH subject authority records have a scope note.

Sense boundary. The categorical distinction between different meanings of a polysemous or homonymous word. In LCSH, sense boundaries are encoded through separate authority records, parenthetical qualifiers, scope notes, or position in the polyhierarchical graph. The binary variable `same_heading` operationalizes sense boundary status in this study.

User warrant. The principle that classification should reflect actual user search terminology and behavior (Beghtol, 1986). User warrant emphasizes lexical preference and cognitive accessibility. Cognitive warrant extends user warrant by targeting the categorical structure of sense boundaries rather than surface terminology and by grounding the evaluation criterion in behavioral evidence rather than user surveys or query log analysis.

CHAPTER 1: INTRODUCTION

Before examining the Library of Congress Subject Headings (LCSH) in detail, it is useful to establish the normative framework within which subject heading languages are designed and evaluated. The International Federation of Library Associations and Institutions (IFLA, 1999) defines the relevant terms as follows:

> Subject heading: a linguistic expression (a word or group of words) representing the subject content of a document and used for retrieval in a [library catalog]...
>
> Subject Heading Language [SHL]: a documentary language used to provide consistent access in a [library catalog] to the subject content of documents... It consists of a controlled vocabulary of terms representing concepts and named entities and a semantic structure showing paradigmatic relationships [i.e., synonyms and hyponyms] among these...
>
> Principles: ...fundamental postulates that can guide the construction and application of any SHL whose objective is to further precision and recall in retrieval. (p. 1)

These definitions already expose a tension. A subject heading language is defined simultaneously as a representational structure—a controlled vocabulary organizing concepts and their relationships—and as an instrument of retrieval, evaluated against the operational criteria of precision and recall. This dual character reflects a deeper ambiguity in the relationship between knowledge organization systems (KOS) and information retrieval (IR) that has become more pronounced as library discovery systems have grown in complexity.

The concept of *cognitive warrant* proposed in this study sits precisely at this boundary. It is a KOS evaluation criterion—a principle that governs how subject heading distinctions should be drawn—but it is grounded in a theory of semantic processing that belongs equally to the IR research tradition. Cognitive warrant holds that the KOS must be evaluated against criteria derived from how retrieval actually works in human cognition, not only in the algorithm. The representational and operational questions, in other words, cannot be answered independently of one another.

The costs of misalignment are borne by users, propagating across every stage of the search interaction.[1] These failure modes are not uniform but depend on the specific semantic properties of the concepts involved — degree of polysemy or homonymy, hierarchical distance between corresponding LCSH headings, and whether continuous semantic similarity predicts or diverges from categorical assignment — and specifying where, for whom, and under what conditions misalignment occurs requires the level of empirical precision existing evaluation frameworks do not provide.

1.1. Statement of the Problem

Knowledge organization systems such as LCSH play a foundational role in the organization, retrieval, and interpretation of information. These systems encode categorical distinctions that structure how resources are described, discovered, and related. Historically, the development and evaluation of such systems have been guided

[1] These are further covered in Chapter 2

by principles such as literary warrant, whereby categories are justified based on their occurrence and usage within the existing literature. While this approach ensures alignment with documented discourse, it does not guarantee that the resulting conceptual distinctions correspond to how users cognitively represent and process meaning.

A critical but underexamined issue is whether the categorical boundaries encoded in controlled vocabularies reflect empirically observable patterns in human semantic cognition. Knowledge organization systems routinely distinguish between different senses of ambiguous terms—distinctions that may correspond to homonymy or polysemy. However, the extent to which these distinctions align with how individuals interpret meaning in context remains unclear. If the boundaries imposed by a KOS diverge from cognitive representations, this misalignment may have implications for information retrieval effectiveness, user comprehension, and the interoperability of knowledge systems with emerging computational models.

At the same time, advances in computational linguistics and psycholinguistics have made it possible to model semantic similarity and sense differentiation using both human behavioral data and vector-based representations of language. Trott and Bergen (2023) demonstrate that word meaning exhibits both continuous and categorical structure, suggesting that semantic boundaries may be empirically measurable rather than purely conventional. Despite these developments, there has been limited integration of such empirical methods into the evaluation of knowledge organization systems.

The problem, therefore, is a lack of theoretical and empirical grounding for assessing whether the semantic distinctions encoded in systems like LCSH correspond to human cognitive processing of meaning. Without such grounding, it remains uncertain whether existing subject heading structures optimally support user interaction or whether they reflect historically contingent categorizations that may not align with contemporary understandings of semantic structure.

1.2. Purpose of the Study

The purpose of this study is to develop and empirically evaluate the concept of cognitive warrant as a principle for knowledge organization, defined as the extent to which the categorical and structural distinctions encoded in a knowledge organization system correspond to observable patterns in human semantic processing.

To achieve this goal, the study integrates methods from library and information science, computational linguistics, graph theory, information theory, and psycholinguistics. It utilizes a dataset of ambiguous word senses drawn from experimental research on semantic processing (Trott & Bergen, 2023), including contextualized sentence stimuli designed to activate distinct meanings of polysemous and homonymous terms. Each word sense is systematically mapped to a LC subject heading through an expert-assisted adjudication process. The study constructs a computational representation of LCSH as a graph-based knowledge structure derived from its bibliographic data.[2]

Using these mappings, the study calculates structural variables characterizing the relationship between subject headings corresponding to different senses of the same word, and then compares these structural measures to empirical indicators of semantic similarity and processing difficulty derived from experimental data. By examining the correspondence between LCSH-based structural distinctions and

[2]Using Metadata Authority Description Schema/Resource Description Framework (MADS/RDF) files from the LC Linked Data Service (id.loc.gov)

cognitive-semantic patterns, the study assesses whether the organization of subject headings reflects meaningful cognitive boundaries.

1.3. Research Questions

The critical question is not whether cognition is graded and KOS are categorical, but whether professionally imposed categorical boundaries correspond to reliable inflection points in behavioral responses to continuous transitions within semantic space. The present study evaluates this alignment empirically.

If professional categories align with latent cognitive structure, then categorical boundaries should correspond to measurable discontinuities in processing. If they do not, professional structure may introduce cognitive friction during user interaction with the search interface. Cognitive warrant, therefore, becomes a falsifiable claim: that professionally imposed boundaries track psychologically meaningful partitions in semantic space.

1.3.1. Main Research Question

> **RQ0.** *LCSH as a representation of semantic structure*
> To what extent can knowledge organization systems such as LCSH be evaluated as representations of semantic structure using computational semantic similarity measures and behavioral validation?

1.3.2. Supporting Research Questions

> **RQ1.** *Representational structure*
> Do continuous semantic similarity and LCSH categorical boundaries independently predict cognitive processing difficulty (Experiment 1) and heading discrimination judgments (Experiment 2) of lexically ambiguous terms?

Trott and Bergen's primed sensibility judgment paradigm showed that both semantic distance and dictionary-derived sense boundaries independently predict cognitive processing difficulty. RQ1 tests whether the same additive structure holds when LCSH-derived sense boundaries substitute for dictionary-derived ones.

> **RQ2.** *Continuous–categorical interaction*
> Does the magnitude of LCSH categorical boundary effects vary nonlinearly across the continuum of graded semantic similarity during implicit semantic processing (Experiment 1), such that boundary-related processing costs are amplified at intermediate semantic distances where continuous similarity is least diagnostic?

This research question concerns disambiguation load: when graded similarity cues are least diagnostic—providing relatively weak evidence about which meaning is intended—categorical boundary information may exert its greatest behavioral impact. It is intuitive that, at small distances, both word senses are clearly similar regardless of categorical assignment (e.g., "He saw a wooden bat" vs. "He saw a baseball bat"; `distance_bert` ≈ 0.098) and that, at large distances, both senses are clearly dissimilar (e.g., "She had a toy block" vs. "She had a writer's block"; `distance_bert` ≈ 0.75). The critical "ambiguous middle range" is where users need categorical information to disambiguate—where semantic distance alone leaves substantial uncertainty about whether two contexts refer to the same or different word senses (e.g., "He was in a distinguished position" vs. "He was in a yoga position"; `distance_bert` ≈ 0.38). The central question is therefore whether the magnitude of the LCSH boundary penalty peaks at intermediate semantic distances and attenuates at both extremes. This is not simply another distance $\times$ boundary interaction test; it is a theory of when categorical structure becomes cognitively consequential.

The distinction from Trott and Bergen's (2023) approach is worth stating precisely. They asked: does the effect of semantic distance on response time *differ* across dictionary sense boundaries—that is, do same-sense and different-sense pairs show different linear slopes? They found the additive model fit better, meaning the boundary shifts the intercept but not the slope. The present question is different in kind: when continuous similarity is indeterminate, does the cognitive system rely more heavily on categorical boundaries to resolve ambiguity? That is a question about *conditional boundary magnitude*, not about slope differences. Their linear interaction specification (`distance_bert` × `same`) tests whether the slope of the distance effect differs across the boundary—it cannot detect a non-linear modulation of boundary magnitude.

It is important to note that, even if RQ2 is answered in the affirmative, Trott and Bergen's additive model can still be the correct general description of how meaning is processed. The nonlinear amplification predicted here is not a refutation of their findings but a refinement: the study tests whether professional categorical distinctions are most cognitively consequential specifically when graded similarity cues are insufficient to resolve ambiguity. If supported, this would provide both theoretical precision about the conditions under which discrete categories have disproportionate cognitive effects and practical guidance for controlled vocabulary editors about where boundary clarification yields the greatest benefit.

> **RQ3.** *Structural refinement*
> Do structural properties of LCSH—specifically hierarchical distance and ambiguity type—systematically refine or explain variation in categorical boundary effects, and, if so, does hierarchical distance mediate any observed moderation by ambiguity type?

Hierarchical distance. LCSH adds a hierarchical dimension absent from the dictionary structure used by Trott and Bergen. Dictionaries organize meaning through headwords and definitions (word senses), where the only structural relationship is whether definitions share the same headword. Proximity in the dictionary provides no information about semantic relationships between entries. LCSH, by contrast, organizes subject headings into polyhierarchical relationships through broader/narrower term links (BT/NT), which can be represented as a directed graph where headings can have multiple broader terms.

Hierarchical distance between two subject headings measures the combined shortest path length from each heading to the pair's lowest common ancestor within the LCSH graph. Unlike vector embeddings produced by BERT, where semantic similarity varies continuously in high-dimensional space, graph distances are discrete steps through connecting relationships. This study's representation of LCSH treats each BT/NT link as equivalent (i.e., uniform edge weighting) for path length calculations, regardless of how semantically close concepts might be in natural language usage.

This research question addresses whether professional catalogers' hierarchical organizational judgments capture cognitively meaningful relationships beyond what distributional semantics already represent. LCSH hierarchy reflects explicit editorial decisions based on knowledge organization principles ("How are these concepts organizationally related?"), while BERT embeddings capture implicit relationships from natural language co-occurrence ("How similarly are these terms used in context?"). If hierarchical distance significantly predicts processing difficulty after controlling for BERT semantic distance, this would validate LCSH structure as cognitively warranted rather than merely organizationally convenient—with implications for whether preserving hierarchical relationships in library discovery systems provides genuine cognitive benefit to users.

Ambiguity type. By definition, homonymous senses are historically unrelated — their surface-form identity is etymological accident rather than semantic derivation. Consequently, each sense maps to an LC subject heading on the basis of its own semantic domain, without structural constraint from the other sense's mapping. The two headings therefore tend to fall in entirely separate hierarchical branches, producing large `hier_distance` values as a predictable structural consequence of homonymy. The categorical boundary and the hierarchical gap are mutually reinforcing: every homonymous boundary-crossing in LCSH tends to be both a formal sense discontinuity and a large structural gap.

Polysemous senses, by contrast, are semantically derived from a common root meaning. Their sense-to-heading mappings are therefore not independent: the shared semantic ancestry tends to constrain both headings to the same or nearby hierarchical regions, even when they fall on opposite sides of a formal categorical boundary. A polysemous boundary-crossing may thus be categorically real while remaining hierarchically proximate.

This structural asymmetry generates the H5 prediction: boundary effects should be stronger for homonymous items than for polysemous ones. Critically, however, this is a *structural* prediction rather than a *cognitive* one — it is grounded in the expected distribution of `hier_distance` values across ambiguity types, not in any claim that the cognitive system processes homonymy and polysemy differently at the categorical boundary. H6 disentangles the two accounts. If the ambiguity type moderation is entirely structural — driven by the larger hierarchical distances systematically associated with homonymous boundary-crossings — then the interaction between ambiguity type and `same_heading` should attenuate substantially or disappear once `hier_distance` is entered as a covariate. If the interaction instead survives this control, it would indicate a genuinely cognitive distinction between homonymy and polysemy above and beyond what LCSH structure captures.

The theoretical stakes of the two outcomes differ sharply. A confirmed H5 that is fully mediated by H6 (structural account) is *consistent* with Trott and Bergen's well-powered null result on the analogous interaction: the reason ambiguity type did not moderate boundary effects in their dictionary-based paradigm is that dictionaries lack the hierarchical amplification that LCSH introduces. On this account, the cognitive system does not distinguish homonymy from polysemy at the categorical boundary — LCSH structure simply places homonymous pairs further apart, and processing difficulty follows hierarchical distance, not ambiguity type per se. A confirmed H5 that is *not* mediated by H6 (cognitive account) would be in genuine tension with Trott and Bergen's null result, because it would imply a cognitive distinction their well-powered study failed to detect, requiring stronger theoretical justification than the current framework provides.

> **RQ4.** *Cross-measure convergence*
> Do effects observed in implicit processing measures converge with explicit appropriateness judgments, indicating stable semantic alignment across task modalities and operationalizations?

RQ4 evaluates whether cognitive warrant reflects a coherent alignment construct rather than a task-specific artifact. If the same predictors that drive implicit processing costs in Experiment 1 also drive explicit discrimination judgments in Experiment 2—at the item level, not just in direction—this convergence provides multi-method evidence for the cognitive reality of LCSH categorical structure.

1.3.3. Synthesis

The four research questions form an interrelated architecture. RQ1 establishes the foundational claim: independent effects of graded semantic distance and LCSH categorical boundaries across both experiments. RQ2 refines this claim for implicit processing, asking whether boundary effects are nonlinear and amplified at intermediate distances. RQ3 asks whether LCSH's polyhierarchical structure adds explanatory value beyond categorical boundary status, and whether ambiguity type moderates these effects or is itself explained by structural asymmetries in LCSH. RQ4 assesses the construct validity of cognitive warrant.

1.4. Significance of the Study

1.4.1. Significance for Psycholinguistics

This study extends Trott and Bergen's (2023) hybrid meaning framework to a new class of categorical boundary: one imposed by professional indexing practice rather than by lexicographic consensus. Trott and Bergen demonstrated that word meaning exhibits both continuous and categorical structure, with cosine distance between contextualized embeddings and dictionary sense boundaries independently predicting sensibility judgment response times. Their boundary variable was defined by expert lexicographers whose work is explicitly aimed at capturing psychologically real sense distinctions. LCSH boundaries, by contrast, were developed through literary warrant and cataloging tradition—a pragmatic and historically contingent process with no theoretical commitment to cognitive validity.

Testing whether professionally imposed boundaries produce the same processing discontinuities as lexicographic boundaries thus addresses a theoretical question that Trott and Bergen's paradigm leaves open: whether the categorical component of lexical processing reflects a general property of how any consistent, socially sanctioned categorical structure corresponds to the human cognitive system, or whether it is specific to boundary systems that were constructed with cognitive validity as a design criterion. A positive result would suggest that the categorical effect is robust to boundary origin; a null or attenuated result would constrain theories of categorical meaning by linking the effect to the nature of the categorization process. Either result will be informative.

1.4.2. Significance for Interdisciplinary Research

This study is designed as an exercise in disciplinary synthesis. Psycholinguistics, working from its own data sources, has access to dictionary sense boundaries and behavioral measures, and has a long-standing tradition of testing whether taxonomic or hierarchical structure moderates categorical processing — from Collins and Quillian's (1969) network-distance account of category verification, through the typicality-distance debate (Rips et al., 1973), to the use of WordNet-derived hierarchical distance as a predictor of human semantic judgments (Maki et al., 2004; Jiang & Conrath, 1997). What this tradition has not done is ask the question in a form that bears on cataloging validity. It has not evaluated whether a professionally curated, literary-warrant-based classification system like LCSH draws its categorical boundaries in places that are cognitively real. This is a different kind of object than a general-purpose lexical hierarchy such as WordNet, which was built for coverage rather than for encoding cataloger judgment. Nor does this tradition ask whether structural properties specific to that kind of polyhierarchy—hierarchical distance, branch membership, shared ancestry—moderate categorical processing effects in a way that speaks to the vocabulary's own construct validity. Knowledge organization research can describe LCSH structure and document cataloger behavior, but lacks the behavioral measurement tools to test whether its distinctions are cognitively real for non-expert users. The integration of LCSH graph structure, BERT-based semantic distance, and psycholinguistic experimental design produces testable hypotheses that belong to neither discipline alone.

1.5. Assumptions

1. BERT semantic distance is a valid proxy for human semantic similarity at the level of continuous grading, capturing something meaningful about graded semantic structure even if it does not perfectly model all dimensions of human lexical representation.
2. LC Linked Data Service authority records accurately reflect professional cataloging judgment for the heading types examined, and the BT/NT structure of those records reflects real (if imperfect) hierarchical organization.
3. Participants recruited through university Sona Systems represent a reasonable proxy for library discovery system users in terms of basic semantic processing—though not in terms of LCSH familiarity, which is explicitly addressed in exclusion criteria.
4. Native (or native-level) English speakers process polysemous word senses in ways sufficiently systematic to be captured by random effects in mixed-effects models.
5. Trott and Bergen's original stimulus sentences retain ecological validity when recontextualized within an LCSH framework.

6. LCSH categorical boundaries can be operationalized as the discrete binary variable `same_heading`, even though LCSH's polyhierarchical structure means some word senses could plausibly map to multiple headings.

1.6. Limitations

1. Sample representativeness. Recruiting from a university participation pool introduces a convenience sample biased toward educated young adults, limiting generalizability.
2. LCSH granularity mismatch. LCSH is coarser-grained than dictionary senses for common polysemous words; some word senses will collapse to the same heading, compressing variation needed to detect boundary effects.
3. Noisy hierarchical signal. LCSH's BT/NT links were generated programmatically in the 1980s and many remain unreviewed (Young, 2017); `hier_distance` is a noisier proxy for conceptual proximity than a purpose-built thesaurus would yield.
4. Ecological validity. The sensibility judgment and appropriateness rating tasks are laboratory analogues of library discovery behavior; whether effects translate to real-world retrieval behavior remains an open question.
5. Single-session design. The within-subjects paradigm cannot assess test-retest reliability.
6. Pre-coordination simplification. Mapping each sense to one heading simplifies LCSH's actual multi-heading indexing practice.

1.7. Chapter Summary

This chapter has established the motivating problem, theoretical orientation, and research architecture that structure the remainder of the study. Drawing on Trott and Bergen's (2023) hybrid model of lexical meaning, the chapter proposes an integrated research framework that treats LCSH as a structured, empirically testable claim about semantic organization. Four supporting research questions organize this framework across representational, interactional, structural, and convergence tiers. Chapter 2 situates this framework within the relevant literature. Chapter 3 develops the theoretical model and formalizes the hypotheses. Chapter 4 describes the methods. Chapters 5 concludes.

CHAPTER 2: REVIEW OF THE LITERATURE

This chapter establishes why cognitive warrant matters to more than just those studying controlled vocabularies. Three claims do that work. First, LCSH's categorical boundaries are invisibly embedded in ordinary keyword search—not confined to a subject-search box most patrons never open. Second, cognitive misalignment propagates across all stages of a user's subject search. Finally, despite roughly fifty years of documentation that LCSH's categories and users' categories diverge, that evidence has never been assembled into a structural account of *why* the divergence occurs or a method for measuring *where* it occurs — the gap this study's framework fills.

Mutual information as a quantity governing the magnitude of category effects in a lexical-semantic system has precedence in the categorical-perception and efficient-communication literature, but that review is deferred to Section 3.4. The discussion of previous cognitive science research with information-theoretic accounts of perceptual learning and lexical semantic typology grounded in the information bottleneck principle will be more fruitful after the theoretical framework of cognitive warrant has been established in Sections 3.1 and 3.2.

2.1. The Different Functions of Dictionaries and LCSH

I want to first note the distinction between dictionaries and LCSH as knowledge organization systems. Certainly, a word or concept can be defined using the categories it belongs to and the meaning of words and concepts are used in their categorization, yet dictionaries and LCSH can be distinguished by their epistemic aims of the organization of meaning and "aboutness," respectively. Or put another way, what something *is* versus how it can be *categorized*.

Even within these two, the meaning of a word and an entity's categorization can each be done in numerous, if not unlimited, ways. Findings over the last forty years establish that controlled vocabulary application involves discretionary professional judgment that does not uniformly reflect either user cognition or consistent professional consensus (e.g., Furnas et al., 1987; Monreal & Gil-Leiva, 2011). On the other hand, Assefa (2007) demonstrated the similarity and predictability of participants' categorizations of 60 different food names through a combination of algorithmic modeling, multidimensional scaling, and text analysis.

I avoid here a discussion of the ontology of meanings and categories. My intent is, if anything, to acknowledge the different functions of dictionaries and LCSH so that it is clear I am not equating them as knowledge organization systems.

2.2. LCSH in Library Discovery Systems

2.2.1. Why This Matters, Even to Users Who Never See a Subject Heading

Cognitive alignment is not a niche concern for patrons who deliberately run a subject search — it is universal, because LCSH's categorical boundaries are woven into the ranking, indexing, and refinement logic behind *every* keyword query, including from patrons who believe they are using a plain, Google-like search box. If LCSH's boundaries match how users actually partition meaning, this invisible mediation is a quiet asset: it disambiguates, connects synonyms, and surfaces material a raw keyword string would miss. If, however, the boundaries diverge from users' cognitive boundaries — the empirical question this study tests — the same mediation becomes a quiet liability, down-ranking or hiding relevant results in a way no patron can detect, because nothing in the interface signals a controlled vocabulary was involved.

2.2.2. Keyword vs. Subject Search

Keyword search matches terms literally across all bibliographic fields without controlled-vocabulary mediation, in principle. Users overwhelmingly prefer it — a 14:1 ratio in one academic library study (Library of Congress, 2008) — because it mimics familiar web search and requires no knowledge of authorized terms. But this preference rests on an assumption that does not hold: keyword search is not vocabulary-neutral, and most users cannot tell why their searches fail.

Subject search queries LCSH fields directly, using authorized headings with pre-coordinate syntax, hierarchical relationships, and synonym control via "USE" references. Catalogers assign headings based on "aboutness" judgments meant to capture conceptual content regardless of the author's own terminology.

Modern discovery systems use algorithms such as BM25F and DisMax that blend both methods through hybrid ranking: keyword queries search all fields but boost relevance scoring of records where LCSH headings match, so LCSH-tagged records disproportionately populate the top of results lists, where most users stop.

2.2.3. LCSH Working Behind the Scenes

> **Abridged:** LCSH sits inside the keyword index as the vocabulary behind query expansion, silently supplying synonyms and broader/narrower terms. Gross, Taylor, and Joudrey (2015) found 27–36% of searches successful through LCSH subject search would have failed with keyword-only approaches — demonstrating hidden value, but the same mechanism can deliver hidden harm depending entirely on whether LCSH's structure aligns with how the user conceptualizes the topic.

No "subject search" button is selected anywhere in this interaction, and nothing tells the user a controlled vocabulary shaped their results. Whether that shaping helps or misleads depends on whether LCSH's boundaries are cognitively valid — the reason cognitive warrant matters for every catalog user, not only the fraction who click "subject search."

2.3. The Cascading Effects of Cognitive Misalignment Throughout the Search Process

Abridged: Having established that LCSH mediates ordinary keyword search, the next question is what happens when its categories misalign with a user's own. The answer is not a single point of failure but a chain reaction. Search-interaction models decompose subject searching into roughly five stages: conceptualizing the need, formulating a query, executing the match, examining results, and reformulating. LCSH-user misalignment doesn't hit at one point — it compounds, because each stage feeds the next a degraded input. The pattern that emerges from the empirical literature is that the mismatch shows up first as a *representational* problem, converts into a *lexical* problem at query time, becomes a *retrieval* problem at matching, then a *comprehension* problem at result examination, and finally a *repair* problem the user is not equipped to solve.

The mismatch is a Stage-1 problem masquerading as five separate ones — which is why interventions targeted at any single downstream stage (better cross-references, better authority displays, keyword augmentation) produce the modest, disappointing gains the literature has repeatedly documented. Fixing Stage 3 or Stage 4 cannot repair a Stage 1 boundary error; only knowing *where* the boundaries themselves diverge from cognition — the diagnostic this study's methodology provides — can tell us which categories need structural revision rather than better user interface design.

2.4. Research Gap

> **Abridged:** Fifty years of empirical work document the LCSH–user mismatch. The phenomenon is well recognized; what's missing is treatment of it as a unified object of study rather than fragments admitting only small technical fixes.
>
> Flowers and Cooke (1992) argued that most research had focused on LCSH *vocabulary* but neglected the way its *structure* interacts with the user's knowledge structure — and that this structural interaction is a substantial part of the subject-search problem.
>
> Existing work has enumerated vocabulary shortfalls, targeted single stages, quantified retrieval consequences, and studied cataloger and user mental models independently, but has not produced a structural account of the mismatch, a propagation model, treating the misalignment as an interaction between cataloger and user, mediated by a vocabulary that constrains what each side can express to the other, or an empirical extension to the discovery-layer environment.

2.5. Chapter Summary

Misaligned categorical boundaries compound across every stage of the subject-search process rather than failing quietly at one, yet LCSH's boundaries are load-bearing infrastructure inside ordinary keyword search, not a peripheral subject-search feature. The literature documenting this misalignment, though extensive, remains fragmented; it has never been assembled into a structural account of why LCSH and human categorization diverge or a method for measuring where. The cognitive warrant framework developed in the next chapter responds to that composite gap directly.

CHAPTER 3: THEORETICAL FRAMEWORK

This chapter develops the paper's central conceptual contribution. It generates the study's empirical logic and proceeds through several movements: from the limits of literary and user warrant to the definition of cognitive warrant (Section 3.1); to an information-theoretic restatement of cognitive warrant as uncertainty reduction (Section 3.2) and the mathematical entailment of H3 (Section 3.3); to a review of the literature involving information theory in cognitive science (Section 3.4); to metrics derived from cognitive information (Section 3.5); to addressing the graduation problem identified by Trott and Parkinson-Coombs (2026) (Section 3.6); and finally to the formal hypothesis set (Section 3.7).

3.1. From Literary Warrant to Cognitive Warrant

3.1.1. Limits of Literary and User Warrant

> **Abridged:** Literary warrant holds that classification systems should reflect the published record: concepts warrant headings when sufficient literature accumulates on them. Literary warrant, however, is insensitive to cognitive plausibility: a subject heading can have strong literary warrant while encoding a category that users find incoherent. Furthermore, a cataloger assigns a heading based on what a document is about—a professional judgment about category membership—without any mechanism for verifying that the resulting category corresponds to a cognitively natural partition.
>
> User warrant broadens evaluation by incorporating actual user search terminology and behavior. Yet user warrant research typically asks which *term* users prefer, not

> whether the categorical *structure* of sense distinctions corresponds to how users organize conceptual space.

3.1.2. Definition of Cognitive Warrant

Cognitive warrant is defined as the extent to which the categorical and structural distinctions encoded in a knowledge organization system correspond to empirically observable patterns in human semantic cognition. More precisely:

> The categorical structure of a KOS possesses cognitive warrant to the extent that the distinctions it encodes correspond to measurable discontinuities in human judgments of semantic similarity or cognitive processing difficulty, above and beyond what is predicted by continuous semantic distance alone.

This definition is operationalizable: the categorical structure of a KOS has cognitive warrant if the binary predictor $S \in \{0, 1\}$ significantly predicts behavioral outcomes after controlling for the continuous predictor d. It is falsifiable: the hypothesis that LCSH structure has cognitive warrant can be rejected if S adds no predictive value above and beyond d. And it is specific to the categorical structure being evaluated rather than a programmatic assertion about controlled vocabularies in general.

Cognitive warrant is complementary to literary warrant, not competitive with it. Literary warrant establishes that a heading exists and refers to a documented concept. Cognitive warrant asks whether the categorical placement of that heading—the specific sense distinctions it enforces—is defensible from the perspective of human semantic processing. A vocabulary can have strong literary warrant and weak cognitive warrant (if its categories are well-attested but cognitively arbitrary), strong cognitive warrant and weak literary warrant (theoretically possible for a vocabulary designed from cognitive data), or both.

3.1.3. Core Propositions of Cognitive Warrant

The cognitive warrant concept generates five testable propositions that organize the study's hypothesis set and provide the evaluative framework for interpreting results.

Proposition 1 (Conceptual Boundary Proposition). Conceptual boundaries encoded in a KOS correspond to perceived semantic distinctions: meanings assigned to different KOS categories should be perceived as less semantically similar than meanings within the same category.

Proposition 2 (Boundary Effect Proposition). Categorical boundaries in a KOS produce measurable effects in semantic judgment tasks. These effects manifest as increased processing time (Experiment 1) and reduced appropriateness ratings for cross-boundary heading alternatives (Experiment 2). This proposition underwrites H2 and E2-H2.

Proposition 3 (Continuous–Categorical Interaction Proposition). The influence of categorical boundaries depends on the underlying continuous semantic similarity between meanings. Boundary effects are weakest when senses are extremely similar (same sense nearly certain) or extremely dissimilar (different sense nearly certain), and strongest at intermediate semantic distances where continuous similarity leaves maximal uncertainty. This proposition generates the nonlinear prediction at the core of H3.

Proposition 4 (Structural Refinement Proposition). Hierarchical and relational structures within a KOS modulate the cognitive validity of conceptual boundaries. Two headings that are separately categorized but proximately situated in the BT/NT hierarchy should produce weaker boundary effects than headings in entirely disconnected subtrees. This proposition generates H4–H6.

Proposition 5 (Empirical Evaluability Proposition). The cognitive warrant of a KOS boundary system can be evaluated using experimental and computational measures of semantic cognition. The combination of response time data, explicit appropriateness ratings, and distributional embedding distances provides a methodologically sufficient basis for evaluating whether LCSH categorical boundaries meet the cognitive warrant criterion. This proposition underwrites the entire methodological design.

3.2. Cognitive Warrant as Uncertainty Reduction in Semantic Processing

3.2.1. Continuous Semantic Space and Disambiguation Load

The central question in this study is not whether meaning is continuous or categorical, but *where* categorical structure provides useful information during processing.

Let d denote the semantic distance between two contexts and $Y \in \{0, 1\}$ indicate whether they instantiate the same sense (i.e., $y = 1$ for same sense contexts and $y = 0$ for different-sense contexts). The conditional probability $P(Y \mid d)$ defines a mapping from semantic distance to sense identity. At low distances the mapping is highly predictable toward same-sense identity ($P(y = 1) \approx 1$); at high distances, equally predictable toward different-sense identity ($P(y = 0) \approx 1$). At intermediate distances the mapping becomes uncertain, which can be formalized using **conditional**

entropy[3]:

$$H(Y \mid d) = -\sum_{y \in 0,1} p(y \mid d) \log p(y \mid d)$$

This uncertainty represents the disambiguation load. My claim is that cognitive processing difficulty increases with this uncertainty: higher $H(Y \mid d)$ corresponds to longer response times as the cognitive system labors to resolve which sense is intended.

3.2.2. Categorical Structure as Cognitive Information

We define the binary variable $S \in \{0, 1\}$ as the **heading congruence** indicator: $S = 1$ when two contexts are assigned to the same LC subject heading (**congruent**), and $S = 0$ when they are assigned to different LC subject headings (**incongruent**). Neither value is the marked or active condition — congruence and incongruence are symmetric categorical outcomes of the heading assignment process, distinguished by whether the two contexts fall within or across an LCSH authority record boundary. When it is necessary to refer to the aggregate pattern of heading congruence values across items or distance bins, I will use **categorical congruence structure**.

This yields a refined conditional distribution with associated entropy $H(Y \mid d, S)$. The **cognitive information** provided by categorical congruence structure is defined as:

$$\mathrm{CI}(d) = H(Y \mid d) - H(Y \mid d, S)$$

Formally, $\mathrm{CI}(d)$ quantifies, conditional on semantic distance d, the reduction in uncertainty about sense identity Y afforded by heading congruence S — that is, by (explicitly or implicitly) knowing whether the two contexts are (congruently) assigned to the same subject heading or (incongruently) assigned to different headings.

[3] Conditional entropy measures how much uncertainty remains (on average) about a random variable (in this case Y) once we already know the value of another (in this case d). The logarithmic base is 2 and the resulting units of information are called binary digits, or *bits*.

The formulation of cognitive information can be expanded to show the significance of conditional probabilities in its calculation.

$$\mathrm{CI}(d) = \underbrace{-\sum_{y \in \{0,1\}} p(y \mid d) \log p(y \mid d)}_{H(Y|d)} + \underbrace{\sum_{s \in \{0,1\}} p(s \mid d) \sum_{y \in \{0,1\}} p(y \mid d, s) \log p(y \mid d, s)}_{-H(Y|d,S)}$$

This reduction in uncertainty should manifest behaviorally as a reduction in response time:

$$\mathrm{CI}_{RT}(d) \propto \mathbb{E}[\mathrm{RT} \mid d] - \mathbb{E}[\mathrm{RT} \mid d, S]$$

When two words are a certain semantic distance apart, knowing they share a heading should make it faster to judge that they share a meaning—and knowing that they belong to different subject headings should make it faster to judge that they have different meanings. The formula simply says that this speed advantage is the behavioral signature of cognitive information: the difference between how long it takes to respond when you only know how semantically distant two items are, versus how long it takes when you also know whether or not they map to the same subject heading.

Categorical congruence structure is informational to the extent that it reduces processing time by resolving ambiguity. Cognitive warrant is therefore reinterpreted as: *the extent to which a knowledge organization system reduces cognitive processing difficulty by resolving uncertainty in semantic interpretation.*

3.2.3. The Central Prediction

This framework yields a central prediction: the effect of categorical congruence structure on response time will be non-linear across semantic distance, with maximal impact at intermediate distances where $H(Y \mid d)$ is maximized. At low and high distances, uncertainty is already low and categorical information provides little additional benefit. At intermediate distances—the "ambiguous zone"—the categorical congruence structure should produce the greatest reduction in processing time. This prediction aligns exactly with H3 and is the information-theoretic basis for the nonlinear form.

3.2.4. Information-Theoretic Foundations

The formal derivations developed in the subsections that follow require a modest extension of the entropy framework introduced above.

Mutual information $I(X;Y)$ quantifies the reduction in uncertainty about one variable achieved by observing the other and is symmetric:

$$I(X;Y) = H(Y) - H(Y \mid X) = H(X) - H(X \mid Y)$$

Completely unrelated variables possess no mutual information; two variables that are strongly related will have high mutual information.

Conditional mutual information $I(X;Y \mid Z)$ measures how much knowing one variable reduces uncertainty about the other given that a third variable is already known:

$$I(X;Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z) = H(X \mid Z) - H(X \mid Y, Z)$$

The Identification: $\mathrm{CI}(d) \equiv I(Y;\, S \mid d)$

The cognitive information quantity defined above is immediately recognizable as conditional mutual information:

$$\mathrm{CI}(d) = H(Y \mid d) - H(Y \mid d, S) \;\equiv\; I(Y;\, S \mid d)$$

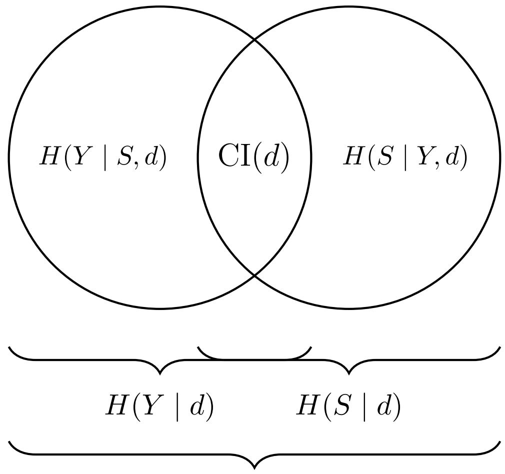
This identification is the master key to the derivations that follow. Every theorem proven about $I(X;Y \mid Z)$ applies directly to $\mathrm{CI}(d)$, and every informational identity translates into a claim about cognitive warrant.

Two corollaries are immediate. First, because conditional mutual information is always non-negative, $\mathrm{CI}(d) \geq 0$ unconditionally: LCSH categorical structure cannot, *on average*, increase uncertainty about sense identity at any given distance (see Figure 3.1). Second, the chain rule applied to the three-variable system (Y, d, S) yields:

$$\begin{aligned} I(Y;\, d, S) &= H(Y) - H(Y \mid d, S) \\ &= H(Y) - \big[H(Y \mid d) - I(Y;\, S \mid d)\big] \\ &= \big[H(Y) - H(Y \mid d)\big] + I(Y;\, S \mid d) \\ &= \underbrace{I(Y;\, d)}_{\text{continuous contribution}} + \underbrace{I(Y;\, S \mid d)}_{\mathrm{CI}(d):\ \text{categorical contribution}} \end{aligned}$$

Total information about sense identity from the complete model decomposes into the contribution of continuous distance alone plus the additional contribution of the categorical congruence structure conditional on distance. The alignment with Trott and Bergen's (2023) hybrid meaning theory is clear.

The null hypothesis for H2 has an exact information-theoretic restatement. Without conditioning on anything, sense identity Y and heading congruence S are dependent by construction: knowing that two contexts map to the same subject heading shifts the probability that they express the same word sense. But this raw dependence is uninformative, because d is a common predictor of both variables—semantically near pairs are simultaneously more likely to be the same sense *and* more



$$H(Y, S \mid d) = H(Y \mid d) + H(S \mid d) - I(Y; S \mid d)$$

$$\boxed{\Rightarrow\ 0 \leq \mathrm{CI}(d) = H(Y \mid d) + H(S \mid d) - H(Y, S \mid d) \leq \min(H(Y \mid d), H(S \mid d))}$$

Figure 3.1: Entropy and mutual information (all conditioned on d) as overlapping regions of $H(Y \mid d)$ and $H(S \mid d)$. $\mathrm{CI}(d) \equiv I(Y; S \mid d)$ is the shared overlap — uncertainty that either variable resolves about the other, conditioned on d. The conditional entropies $H(Y \mid S, d)$ and $H(S \mid Y, d)$ are the portions exclusive to each variable. The conditional joint entropy $H(Y, S \mid d)$ is the total area of both circles combined. From the figure, we can informally conclude that $\mathrm{CI}(d)$ is non-negative and no greater than either $H(Y \mid d)$ or $H(S \mid d)$. This is important for warrant efficiency (Section 3.5.1) and the redundancy bound (Section 3.5.6).

likely to reside in the same heading. The dependence between Y and S may therefore be entirely mediated by their shared relationship with d. In the null hypothesis, Y and S are conditionally independent given d.

The study's central question—whether LCSH categorical structure S adds predictive value above and beyond semantic distance d—is equivalent to asking whether this conditional independence is violated. In information-theoretic terms, this is precisely the question of whether $\mathrm{CI}(d) = I(Y; S \mid d) > 0$.

If $\mathrm{CI}(d) = 0$, heading congruence is fully redundant with BERT distance: a sufficiently fine-grained continuous similarity measure renders the categorical partition cognitively unnecessary. If $\mathrm{CI}(d) > 0$, LCSH categories carve semantic space in a way the continuous geometry cannot fully capture—the categorical congruence structure has independent cognitive reality beyond what distributional similarity encodes.

3.2.5. Convergent Validity from Surprisal: Trott and Bergen's Supplementary Analysis

The identification of $\mathrm{CI}(d)$ as conditional mutual information rests on treating d (operationalized as `distance_bert`) as the continuous covariate against which the categorical contribution of S is assessed. Cosine distance, however, is a geometric quantity—a measure of angular separation between two embedding vectors—and is not itself derived from a probability distribution. This raises a legitimate question for the validity of the cognitive information construct: is the categorical boundary effect doing genuine information-theoretic work, or is it merely compensating for the limitations of cosine distance as a proxy for uncertainty? A stronger test would replace the geometric covariate with one that is itself an explicit Shannon quantity, and ask whether the categorical effect survives.

Trott & Bergen (2023) conducted exactly this test in their Supplementary Analysis 3. Using BERT-large, they computed **surprisal**—the negative log-probability the model assigns to the critical word given the preceding sentential context:

$$\mathrm{Surprisal}(w_t) = -\log P\big(w_t \mid w_1, \ldots, w_{t-1}\big) \tag{3.1}$$

Unlike cosine distance, Equation 3.1 is self-information in the literal Shannon sense: it quantifies the uncertainty resolved by observing the target word, conditional on context. Trott & Bergen fit a full model with fixed effects of Surprisal, Cosine Distance, Sense Boundary, and Prior Accuracy (or RT), with random intercepts for participants and words, and compared it against reduced models omitting each BERT-derived term in turn.

Two results from this analysis bear directly on the validity of CI(d) as a construct.

First, Sense Boundary remained predictive of both dependent measures even after Surprisal was included alongside Cosine Distance—for Accuracy, $\chi^2(1) = 188.66$, $p < .001$; for Log RT, $\chi^2(1) = 71.58$, $p < .001$. This is a more stringent test than the primary analyses reported in the main text, because Surprisal is not merely another embedding-derived distance but an explicit probabilistic quantity: as $-\log P(w_t \mid w_1, \ldots, w_{t-1})$, it is the model's own estimate of the uncertainty resolved by observing the target word, with an established independent literature connecting it to processing cost (Levy, 2008). That the categorical effect survives this test indicates the boundary effect is not an artifact of Cosine Distance specifically being an inadequate uncertainty proxy; it persists against the best available continuous information-theoretic competitor, not merely against a geometric one.

For the present framework, this constitutes precedent evidence of a precise formal relationship. Recall that CI(d) $\equiv I(Y; S \mid d)$, which expands as $H(Y \mid d) - H(Y \mid d, S)$: the reduction in uncertainty about Y produced by knowing S, above and beyond what d already captures. CI(d) > 0 therefore holds if and only if S carries information about behavioral outcomes that is irreducible to the continuous distance predictor. In Trott and Bergen's paradigm, Sense Boundary plays the role of S and Surprisal plays the role of d; demonstrating that Sense Boundary remains predictive after Surprisal is controlled is precisely demonstrating that $H(Y \mid \text{Surprisal}) > H(Y \mid \text{Sense Boundary, Surprisal})$—the entropy inequality that

$\mathrm{CI}(d) > 0$ asserts. Because Surprisal operationalizes the model's own probabilistic uncertainty rather than a geometric proxy of it, this irreducibility cannot be attributed to d's failure to measure uncertainty adequately. It is instead evidence that a categorical boundary variable encodes something about processing difficulty that no continuous predictive uncertainty measure, however well-calibrated, fully captures.

The "precedent" qualifier signals a structural dissimilarity that must be kept in view. Trott and Bergen's S is a dictionary sense categorization and their continuous competitor in the Supplementary Analysis is Cosine Distance *and* Surprisal jointly; the present framework uses `same_heading` as S and `distance_bert` alone as d, making the formal target $I(Y; \texttt{same\_heading} \mid \texttt{distance\_bert}) > 0$, a related but strictly distinct claim. What Supplementary Analysis 3 licenses is the inference that this class of irreducibility is empirically achievable in a closely analogous paradigm, raising the antecedent plausibility of H2 without constituting a direct test of it. Notably, Trott and Bergen's result is formally *stronger* than what H2 requires: they demonstrate irreducibility against both continuous predictors jointly, whereas the present hypothesis conditions on `distance_bert` alone. Their precedent therefore excludes a broader class of reductionist alternatives than $\mathrm{CI}(d) > 0$ strictly demands—making it a conservative rather than merely analogous foundation for the present claim.

Second, Surprisal itself explained variance independent from both Cosine Distance and Sense Boundary, for Accuracy [$\chi^2(1) = 13.57$, $p = .001$] and RT [$\chi^2(1) = 27.36$, $p < .001$], with higher Surprisal predicting lower Accuracy ($\beta = -0.05$, $SE = 0.01$) and longer RT ($\beta = 0.19$, $SE = 0.02$). Notably, Surprisal predicted *both* dependent measures, whereas Cosine Distance predicted RT only. Insofar as Surprisal is a probability-based quantity and Cosine Distance is not, this asymmetry is consistent with—though does not by itself establish—the broader claim underlying CI(d): that uncertainty-resolution quantities, rather than geometric distance per se, are the more behaviorally diagnostic currency for modeling processing difficulty, including binary outcomes like response accuracy that geometric distance alone failed to predict in the primary analysis.

Read together, these findings function as convergent, precedent-level validity evidence for the theoretical claim that categorical congruence structure is not redundant with continuous distributional measures, even when those measures are upgraded from geometric distance to genuine Shannon surprisal. This bears on the present study's central wager: that `same_heading` will likewise carry information about Y beyond what `distance_bert` captures. The redundancy bound $\mathrm{CI}(d) \leq H(S \mid d)$ established in Section 3.5.6 caps this contribution by the residual uncertainty in S once d is already known; the bound inhibits—forcing CI(d) toward zero by construction, independent of any real cognitive effect—only to the extent that `same_heading` is itself well-predicted by `distance_bert`. By analogy, Trott and Bergen's own Cosine Distance and Sense Boundary variables were only moderately associated ($r_{pb} = -0.505$, 95% CI $[-0.546, -0.463]$, $p < .001$, leaving 74.5% of BERT-distance variance unexplained by boundary status), which is consistent with an indefinite bound in their design. It is plausible, by the same logic, that an equivalent moderate, non-collinear relationship will exist between `same_heading` and `distance_bert` in the present stimuli. This is a precedent-based plausibility argument, however, not yet an empirical finding about the present design: whether the bound actually binds here depends on the `same_heading`–`distance_bert` relationship within the LCSH mapping, which can only be assessed once that relationship is computed from the stimulus set.

A Necessary Caveat. This evidence should not be overstated. Surprisal as computed in Supplementary Analysis 3 is a fundamentally different construct from the pairwise semantic distance that CI(d) conditions on. It is an asymmetric measure of *context-dependent lexical predictability*—how unexpected a specific wordform is given the immediately preceding sentence—not a symmetric measure of conceptual separation between two senses or contexts of use. Two sentence pairs could differ substantially in cosine distance while showing similar surprisal (if both target words are locally predictable from their respective contexts despite expressing different senses), or vice versa. Surprisal and cosine distance are correlated operationalizations of contextual fit, not interchangeable measures of the same underlying quantity, and the asymmetry in what each predicts (Accuracy and RT versus RT alone) may partly reflect this difference in what is being measured rather than a clean demonstration that information-theoretic quantities are categorically superior to geometric ones.

This caveat is reinforced by Trott and Bergen's own finding, reported earlier in the same set of supplementary analyses, that which embedding-derived measure best predicts which dependent variable is sensitive to model architecture and size: BERT-base cosine distance was the strongest RT predictor, while BERT-large cosine distance was the strongest predictor of Correct Response. The pattern of results is therefore plausibly conditioned on the specific architecture queried, not an inherent property of surprisal as a class of measure.

Finally, and most importantly for the present study, this is *convergent* validity from an analogous paradigm—not a direct empirical test of CI(d) on the LCSH stimuli. Experiment 1 does not include a masked-word surprisal task, and no claim is made here that the LCSH design replicates this specific result. What Supplementary Analysis 3 licenses is a narrower and more defensible claim: that the general theoretical move underlying CI(d)—testing whether categorical congruence structure survives the strongest available continuous probabilistic competitor—has prior empirical support in a closely related paradigm, which increases the antecedent plausibility of H2 without substituting for the planned test of CI(d) > 0.

3.3. H3 as a Mathematical Consequence of the CI Framework

In Section 3.2.3, it was stated that H3 is predicted by the cognitive information framework; however, the most consequential implication of the information-theoretic identification of $\mathrm{CI}(d) = I(Y; S \mid d)$ is that H3 is not merely a prediction but a *mathematical consequence* of the framework under a minimal distributional assumption about the stimulus-level generative process.

3.3.1. The Distributional Assumption and Its Justification

Let $P(Y = 1 \mid d) = \sigma(\alpha - \beta d)$, where $\sigma(\cdot)$ is the logistic function and $\alpha, \beta > 0$. This assumption is a theoretical claim about how the probability of a same-sense judgment at the *stimulus level* varies with BERT semantic distance—not a claim about participant responses (which involve judging individual sentence sensibility) and not directly embedded in the linear mixed-effects models for `log_rt`. Its justification is independent of those models and rests on three grounds.

Maximum entropy

A maximum-entropy distribution is, within a specified class of candidates, the one whose entropy is at least as large as that of every other member of the class. The principle of maximum entropy holds that if the only thing known about a distribution is that it belongs to such a class—defined by a small number of constraints rather than a fully specified functional form—then the maximum-entropy member is the appropriate default, since it is the least informative distribution consistent with what is actually known. This choice is usually motivated on two grounds: it minimizes the information built into the distribution beyond what the constraints license (any lower-entropy alternative satisfying the same constraints encodes some additional regularity—curvature, asymmetry, multimodality—creeping in through an unmotivated choice of form), and, in physical systems, entropy-maximizing configurations are often what the system evolves toward on its own. Of course, only the first motivation applies to $P(Y \mid d)$.

Among all conditional distributions consistent with the one fact we are willing to commit to, that Y and d co-vary linearly on average, the maximum-entropy member is the one that assumes nothing further about *how* Y depends on d, whereas any other choice would tacitly assume additional structure unsupported by that fact. Let $\pi(d)$ denote the marginal density of d over the stimulus set, and let $p(d) \equiv P(Y = 1 \mid d)$ be the (unknown) conditional probability function to be determined. The problem is: among all functions $p(d) \in [0, 1]$, find the one that maximizes the average conditional entropy

$$\bar{H} \;=\; \mathbb{E}_d\big[H(Y \mid d)\big] \;=\; \mathbb{E}_d\Big[-p(d)\log p(d) - (1 - p(d))\log(1 - p(d))\Big],$$

subject to two moment constraints fixed by their empirical values:

$$\mathbb{E}[Y] = c_0, \tag{3.2}$$

$$\mathbb{E}[Yd] = c_1. \tag{3.3}$$

Constraint (3.2) fixes only the overall base rate of same-sense judgments across the stimulus set; constraint (3.3) fixes only the degree to which Y and d co-vary linearly, on average, across the set. Neither constraint says anything about the *shape* of $p(d)$

as a function of d—no quadratic term, no threshold, no assumption of monotonicity is imposed. The claim is that these are the only two facts about the (Y, d) relationship that the theory is entitled to assume a priori, because both are directly estimable from the stimulus set without presupposing any particular response function.

Introducing Lagrange multipliers λ_0 for (3.2) and λ_1 for (3.3), the Lagrangian is

$$\mathcal{L} \;=\; \mathbb{E}_d\big[H(p(d))\big] \;-\; \lambda_0\big(\mathbb{E}[Y] - c_0\big) \;-\; \lambda_1\big(\mathbb{E}[Yd] - c_1\big).$$

Because $\bar{H}$ and both constraints are additive over d (weighted by $\pi(d)$), the optimum can be found pointwise: differentiating the integrand with respect to $p(d)$ for each fixed d and setting the result to zero,

$$\frac{\partial}{\partial p(d)}\Big[H(p(d)) - \lambda_0\, p(d) - \lambda_1\, p(d)\, d\Big] \;=\; 0.$$

Since $\partial H(p)/\partial p = \log\big[(1-p)/p\big]$, this gives

$$\log\frac{1-p(d)}{p(d)} \;=\; \lambda_0 + \lambda_1 d \quad\iff\quad \log\frac{p(d)}{1-p(d)} \;=\; -\lambda_0 - \lambda_1 d.$$

Absorbing signs into the multipliers (relabeling $-\lambda_0 \to \lambda_0$, $-\lambda_1 \to \lambda_1$, which is permissible since their values are in any case fixed only implicitly, by (3.2) and (3.3)) yields

$$\log\frac{P(Y=1\mid d)}{P(Y=0\mid d)} \;=\; \lambda_0 + \lambda_1 d, \qquad \text{i.e.} \qquad P(Y=1\mid d) \;=\; \sigma(\lambda_0 + \lambda_1 d),$$

exactly the assumed logistic form, with $\lambda_0 = \alpha$ and $\lambda_1 = -\beta$. The linear log-odds identity is thus not an independent assumption but a *derived consequence* of maximizing entropy subject to nothing more than the empirical base rate and the empirical linear co-variation between Y and d. The values of α and β themselves are pinned down only implicitly, by requiring that $\sigma(\alpha - \beta d)$ reproduce c_0 and c_1 under $\pi(d)$—which are precisely the score equations solved by maximum-likelihood logistic regression. This is not a coincidence: the maximum-entropy distribution subject to a moment constraint on a sufficient statistic and the maximum-likelihood fit of the corresponding exponential-family model are dual solutions to the same optimization problem, so fitting $P(Y = 1 \mid d)$ by logistic regression on the stimulus set *is*, formally, finding the maximum-entropy distribution consistent with c_0 and c_1.

Framed this way, the assumption's minimality becomes precise. The theory does not assume that log-odds are linear in d; it assumes only that Y and d possess a well-defined linear association, of the kind any competent measure of semantic distance should be expected to have with a sense-identity judgment, and that no further structure beyond the base rate and this association is available to constrain $P(Y \mid d)$. Any additional curvature in $P(Y = 1 \mid d)$—a plateau, a sharp threshold, non-monotonicity—would encode information beyond $\mathbb{E}[Yd]$, and by definition the maximum-entropy solution contains no such information unless it is separately justified and imposed as a further constraint.

Monotonicity requirement

The parameter constraint $\beta > 0$ encodes a single behavioral commitment: the probability of a judgment that two stimulus contexts are same-sense is monotonically decreasing in semantic distance. Pairs that are farther apart in BERT space are less likely to be judged as expressing the same sense. Rejecting this would mean that BERT distance is not directionally valid as a measure of semantic similarity—a claim that would undermine the study's measurement framework entirely, not merely H3. The constraint $\alpha > 0$ encodes a second commitment: at $d = 0$, same-sense is more probable than different-sense, i.e., $P(Y = 1 \mid d = 0) = \sigma(\alpha) > 0.5$. Rejecting this would imply that maximally similar contexts are more likely to express different senses than the same sense—an incoherent position. Neither constraint is specific to LCSH or to any claim the study is designed to establish.

Empirical consistency.

Trott and Bergen's (2023) linear mixed-effects model of `log_rt` on `distance_bert` found a significant effect of semantic distance on response time, and their `glmer` model of `correct_response` found a significant effect of distance on accuracy. These results are consistent with the monotonicity requirement $\beta > 0$—not a direct test of the logistic functional form for $P(Y = 1 \mid d)$, since sense identity is a predictor in their models rather than an outcome, but independent empirical support for the directional claim the assumption encodes.

3.3.2. Three Results and the Theorem

Under the logistic assumption, three results follow in sequence.

Result 1: $H(Y \mid d)$ is bell-shaped.

Since Y is binary, its conditional entropy at distance d is fully determined by $P(Y = 1 \mid d)$ via the binary entropy function:

$$\begin{aligned} H(Y \mid d) &= h_b(P(Y = 1 \mid d)) \\ &= -P(Y = 1 \mid d) \log P(Y = 1 \mid d) \; - \; P(Y = 0 \mid d) \log P(Y = 0 \mid d) \end{aligned}$$

The function $h_b(p)$ equals zero at $p \in \{0, 1\}$ and achieves its maximum of $\log 2$ at $p = 0.5$. Tracing what happens as d increases under the logistic assumption: at $d = 0$, $P(Y = 1 \mid d) = \sigma(\alpha) \approx 1$ for large α, so $h_b \approx 0$—same-sense is nearly certain, disambiguation load is minimal. At $d^* = \alpha/\beta$, the logistic equals $\sigma(0) = 0.5$, so $h_b(0.5) = \log 2$—maximum uncertainty, neither sense is favored. As $d \to \infty$, $P(Y = 1 \mid d) \to 0$, so $h_b \to 0$—different-sense is nearly certain, disambiguation load again minimal. Therefore $H(Y \mid d)$ starts near zero, rises to a peak at the distance of maximum ambiguity d^*, and falls back to zero: a unimodal, bell-shaped function of d.

Two precision notes. First, $H(Y \mid d = 0) = 0$ exactly only in the limit $\alpha \to \infty$; for finite α, the entropy at $d = 0$ equals $h_b(\sigma(\alpha)) > 0$, small but nonzero. The qualitative argument is unaffected. Second, although the text refers to $H(Y \mid d)$ as "strictly concave," this characterization is imprecise: a globally strictly concave function cannot equal zero at two points and be positive in between without dipping below zero, which is impossible for an entropy. The correct description is *unimodal*: a single interior maximum with approach to zero at both extremes, possibly with inflection points in the tails.

Result 2: $\mathrm{CI}(d)$ **is squeezed to zero at both extremes.**

Two properties of conditional mutual information jointly constrain $\mathrm{CI}(d)$: non-negativity ($\mathrm{CI}(d) = I(Y; S \mid d) \geq 0$ for all d) and the upper bound ($\mathrm{CI}(d) = H(Y \mid d) - H(Y \mid S, d) \leq H(Y \mid d)$, since $H(Y \mid S, d) \geq 0$). Together:

$$0 \;\leq\; \mathrm{CI}(d) \;\leq\; H(Y \mid d)$$

At both extremes of d, $H(Y \mid d) \to 0$ by Result 1. Since $\mathrm{CI}(d)$ is squeezed between zero and a quantity that goes to zero, it must itself go to zero at both extremes. The interpretation is direct: when sense identity is already nearly certain from distance alone—either because the contexts are very similar or very dissimilar—there is no residual uncertainty for the categorical congruence structure to resolve, and the heading congruence contributes nothing.

Result 3: An interior maximum is forced.

The conclusion follows from continuity. CI(d) is:

1. continuous under the smooth logistic assumption;
2. non-negative everywhere;
3. zero at both extremes; and
4. positive at some interior value of d, if the boundary provides any useful information at all.

A continuous function that is zero at both endpoints and positive somewhere in the interior must achieve its maximum at some interior point. No additional assumption is required. The inverted-U shape is not imposed; it is the only shape consistent with these four properties.

3.3.3. The Theorem and Its Implication for H3

H3 predicts that the boundary effect is largest at intermediate distances. The argument above shows that this pattern is mathematically entailed by the logistic assumption together with the algebraic properties of CI(d)—*neither of which is specific to LCSH.* The only escape route is CI(d) $= 0$ everywhere, meaning that the LCSH categorical congruence structure provides no useful information at any distance. But this is precisely the joint null of H2 and H3. They are therefore logically linked: if the null hypothesis of H2 is rejected—if LCSH categorical congruence structure adds *any* predictive value beyond continuous distance—then the inverted-U shape of H3 follows as a theorem rather than a conjecture. *The regression test of H3 is not asking whether the peak exists in principle; the mathematics guarantees that it must if H2* *holds*. The regression is instead asking whether the peak is large enough to detect at the study's sample size, and where in the distance continuum it falls.

3.4. Information-Theoretic Accounts of Category Structure in Cognitive Science

The formal core of this study's theoretical framework—the claim that a categorical boundary is psychologically real to the extent that it carries conditional mutual information about an outcome variable, above and beyond a continuous covariate—is not without precedent. Two literatures, neither originating in library and information science, have already established that mutual information is a valid and productive lens on category structure: work on categorical perception grounded in rational, information-theoretic accounts of perceptual learning, and work on lexical semantic typology grounded in the information bottleneck principle. Both are reviewed here in some depth, because the credibility of cognitive warrant as a construct depends on showing precisely how it extends these paradigms rather than merely resembling them in vocabulary.

3.4.1. Mutual Information as a Signature of Psychologically Real Boundaries

Feldman (2021) provides the most direct precedent for the information-theoretic logic this study relies on. Participants learned novel categories defined as Gaussian distributions over a continuous two-dimensional shape space; because the generative model was known, Feldman computed the mutual information $I(C; f)$ between category membership and any feature dimension analytically rather than estimating it from behavior. Discrimination along each feature was measured before and after learning, and the central finding was that improvement in discrimination—the classic signature of categorical perception—was proportional to $I(C; f)$: maximally informative features produced the largest perceptual warping, orthogonal features produced none. A separate, independently developed paper (Brasselet & Arleo, 2018), arrives at a convergent conclusion by a different route, deriving categorical perception and internal category structure (prototype and perceptual-magnet effects) jointly from an information-theoretic objective that combines category probabilities with pairwise similarities, rather than assuming category structure as given. Together these two lines of work establish mutual information as a mechanistically validated, not merely rhetorical, construct in the categorical-perception literature: the quantity that predicts the *magnitude* of a category's behavioral footprint, not just its presence—precisely the logical structure the cognitive warrant claim relies on.

Three differences bound the precedent, however. Feldman's categories are freshly learned in a single session under experimenter control, not the product of a century of distributed cataloger judgment. His information quantity is *marginal*, $I(C; f)$, computed because feature and category are independently manipulated; cognitive warrant requires the *conditional* form, $I(Y; S \mid d)$, precisely because d and S are correlated by construction in naturally occurring linguistic material rather than experimentally decoupled—this is why the redundancy bound $\mathrm{CI}(d) \leq H(S \mid d)$ is load-bearing rather than optional. And his dependent measure is low-level perceptual discrimination accuracy, whereas this study's RT and sensibility judgments sit at the level of lexical access. Feldman licenses the general claim that mutual information tracks category reality; it makes no prediction about whether a professionally authored, retrieval-oriented vocabulary will exhibit the same property.

3.4.2. The Information Bottleneck Account of Lexical-Semantic Category Systems

A second, more consequential literature applies information theory directly to lexical-semantic category systems: the efficient-communication program built on the information bottleneck (IB) principle of Tishby et al. (2000) and applied to semantic typology by Zaslavsky et al. (2018a) and Zaslavsky et al. (2018b).

The communication model in Zaslavsky et al. (2018a) formalizes a category system as an encoder $q(w \mid m)$ mapping meanings onto words, evaluated on two competing quantities: complexity $I(M;W)$, how much the system compresses meaning space, and informativeness $I(W;Y)$, how well a listener recovers intended meaning from the word alone. A system is efficient to the extent it sits near the theoretical Pareto frontier of this trade-off. Applied to color naming, this showed that color vocabularies across dozens of unrelated languages sit close to the IB-optimal frontier, that a single trade-off parameter accounts for much cross-linguistic variation, and that near-optimal solutions naturally produce the soft, sometimes inconsistently named boundaries observed empirically—phenomena hard-boundary accounts struggle to explain. The same general efficient-communication logic was shown to hold for kinship terminology by Kemp and Regier (2012), and the IB objective itself was directly re-derived for container and animal categories by Zaslavsky et al. (2019), both with comparable near-optimal fits—making efficient communication under a complexity–accuracy trade-off the dominant formal account of why lexical categories are shaped as they are.

This is the literature closest to the present study's subject matter, and the overlap is genuine: both treat category systems as objects analyzable via mutual information, both take continuous meaning representations as primitive, both ask whether boundaries are principled or arbitrary. A reader familiar with Zaslavsky et al. (2018a) will reasonably ask why cognitive warrant is not simply IB theory applied to LCSH.

The difference is what question each framework answers, and it is structural rather than topical. IB asks an *optimality* question: is the observed system, among all theoretically possible encoders, near the Pareto-optimal frontier—requiring optimization over the space of every alternative way the vocabulary could have been organized? Cognitive warrant asks a *validity* question about one specific, non-evolved system: does the boundary structure LCSH imposes—designed by catalogers under literary and organizational warrant, not shaped by communicative pressure—track a discontinuity in human processing at all, beyond what continuous distributional information already predicts? Formally, IB optimizes the encoder $q(w \mid m)$ itself; $\mathrm{CI}(d) = I(Y; S \mid d)$ instead holds S fixed exactly as LCSH defines it and asks what it adds conditional on a fixed, independently estimated distance d. Because LCSH was never subject to communicative-efficiency pressure, the IB literature offers no basis, on its own terms, for predicting whether $\mathrm{CI}(d) > 0$—this is precisely the gap that motivates an empirical study rather than a purely theoretical one.

3.4.3. Synthesis: What These Paradigms Establish, and What They Leave Untested

Table 3.1 summarizes the comparison developed above. Read together, these two literatures do a great deal of work for this study without pre-empting its central question. Feldman (2021) establishes, with full experimental control, that mutual information between a categorical variable and a candidate predictor is not a metaphorical description of category effects but the quantity that governs their magnitude—licensing the general logical form of the cognitive warrant claim. Zaslavsky et al. (2018a) and Zaslavsky et al. (2018b) establish that this general logic extends specifically to lexical-semantic category systems, and that real human vocabularies can be fruitfully analyzed as information-theoretic objects rather than as arbitrary cultural residue—licensing the application of mutual-information reasoning to word meaning in particular, as opposed to shape or color perception.

Neither literature, however, has asked whether a *professionally authored, retrieval-oriented* controlled vocabulary—one that is not a freshly learned experimental category system and not a communicatively evolved natural-language lexicon, but a third kind of artifact entirely, produced by institutional editorial process under an explicit and documented warrant tradition—exhibits the same information-theoretic signature of psychological reality. Nor has either literature combined the conditional formulation of mutual information with a continuous, contextualized embedding-based covariate in the way the disambiguation-load account of Trott and Bergen (2023) requires, since Feldman's covariate is a hand-designed perceptual dimension and the IB program's covariate is a hand-specified meaning space over color, kin, or object categories rather than a distributional semantic model estimated from usage. Cognitive warrant is best understood, in light of this review, not as an unprecedented formal invention but as the intersection of these two established paradigms with a third literature—the warrant tradition in knowledge organization and the hybrid meaning theory of Trott and Bergen—applied to an object neither the categorical-perception nor the efficient-communication literature has examined: a professional classification system evaluated not for its optimality, but for its cognitive validity.

3.5. Extensions of the CI Framework

The identification $\mathrm{CI}(d) \equiv I(Y; S \mid d)$ opens a family of derivable extensions. Each generates a distinct theoretical claim, empirical diagnostic, or evaluative criterion. They are developed below in order from most direct to most expansive.

	Feldman (2021)	**Zaslavsky et al. (IB)**	**Trott and Bergen (2023)**	**This study**
Category origin	Freshly learned in-lab	Evolved for communicative efficiency	Lexicographic (dictionary senses)	Professionally authored (LCSH, literary/organizational warrant)
Information quantity	Marginal $I(C; f)$	Complexity–accuracy trade-off $I(M; W)$, $I(W; Y)$	Not information-theoretic (HLM)	Conditional $I(Y; S \mid d) = \mathrm{CI}(d)$
Object of study	Perceptual discrimination warping	Optimality of the encoder vs. theoretical frontier	Cont./cat. effect on RT/accuracy	Incremental validity of S given d
Domain	2D shape space (psychophysics)	Color, kinship, artifacts (cross-linguistic)	Word senses (dictionary)	Word senses (LCSH headings)
Relation to this study	Licenses the general MI–reality logic	Licenses info-theoretic treatment of lexical categories; distinguished by validity vs. optimality question	Primary empirical and methodological foundation; boundary source generalized	—

Table 3.1: Positioning cognitive warrant relative to the nearest information-theoretic precedents in categorical perception and lexical semantic typology, alongside the study's direct empirical foundation.

3.5.1. Warrant Efficiency

As shown in Result 2 of Section 3.3, $0 \leq \mathrm{CI}(d) \leq H(Y \mid d)$. Therefore, the maximum cognitive information that categorical congruence structure can provide at distance d is $H(Y \mid d)$. Define the **warrant efficiency** of the heading congruence at semantic distance d as the fraction of disambiguation uncertainty that the categorization resolves:

$$\eta(d) = \frac{\mathrm{CI}(d)}{H(Y \mid d)} = \frac{I(Y; S \mid d)}{H(Y \mid d)}$$

with the convention $\eta(d) = 0$ wherever $H(Y \mid d) = 0$ (i.e., at distances where sense identity is already determinate without categorical assistance). Warrant efficiency ranges from 0 (the heading congruence resolves none of the residual ambiguity at this distance) to 1 (the heading status perfectly resolves all remaining uncertainty about sense identity given the observed distance). Unlike $\mathrm{CI}(d)$, which inherits its scale from $H(Y \mid d)$ and peaks wherever base uncertainty is largest, $\eta(d)$ is normalized and measures *relative* effectiveness: how much of the disambiguation need is actually met by the categorical structure.

The verbal interpretation is direct: the categorical congruence structure of a KOS operates at efficiency η at distance d if it resolves proportion η of the disambiguation load that continuous similarity cannot resolve. A heading congruence with $\eta(d) \approx 1$ at intermediate distances is providing nearly as much categorical information as is theoretically possible given what distributional models already provide. One with $\eta(d) \approx 0$ is cognitively redundant at those distances, regardless of how well the heading assignment is attested in the documentary record (i.e., through literary warrant).

3.5.2. Expected Cognitive Information and the KOS Evaluation Metric

Integrating $\mathrm{CI}(d)$ over the empirical distribution of semantic distances yields a scalar aggregate:

$$\overline{\mathrm{CI}} = \mathbb{E}_d[\mathrm{CI}(d)] = \sum_d \mathrm{CI}(d) \cdot P(d)$$

This is the **expected cognitive information** of the categorical congruence structure—the average uncertainty reduction provided by the structure, weighted by the frequency with which each distance region is encountered in the stimulus set or, for a properly powered full study, in the user population of interest.

$\overline{\mathrm{CI}}$ is a *system-level cognitive warrant score*: a single number summarizing how much cognitive information a KOS provides on average across the full distribution of semantic distances users encounter. It is estimable from behavioral data, additive across independent systems, and directly comparable across different controlled vocabularies evaluated against the same stimulus materials. The comparison $\overline{\mathrm{CI}}_{\mathrm{LCSH}}$ versus $\overline{\mathrm{CI}}_{\mathrm{MeSH}}$ versus $\overline{\mathrm{CI}}_{\text{BERT-cluster}}$ is an empirically resolvable question: which system's categorical structure better tracks the distribution of disambiguation need in the user population?

> **Abridged:** The KOS evaluation literature has historically lacked a quantitative, behaviorally grounded criterion for comparing alternative categorical structures. $\overline{\mathrm{CI}}$ provides exactly this criterion. This reconceptualizes KOS evaluation from a qualitative assessment of consistency with principles to a quantitative assessment of information yield relative to a theoretically principled baseline.

3.5.3. Hierarchical Cognitive Information Increment

The CI framework extends naturally to the hierarchical predictor `hier_distance` (h). Define the **structural cognitive information** as the total uncertainty reduction achieved by the full structural signature of the heading system—heading congruence and hierarchical depth jointly:

$$\mathrm{CI}_{\mathrm{struct}}(d) \;\equiv\; I(Y;\, S, h \mid d) = H(Y \mid d) - H(Y \mid d,\, S,\, h)$$

The chain rule applied to the four-variable system (Y, d, S, h) yields:

$$\begin{aligned}
\mathrm{CI}_{\mathrm{struct}}(d) &= H(Y \mid d) - H(Y \mid d, S, h) \\
&= H(Y \mid d) - \big[H(Y \mid d, S) - I(Y;\, h \mid d, S)\big] \\
&= \big[H(Y \mid d) - H(Y \mid d, S)\big] + I(Y;\, h \mid d, S) \\
&= \underbrace{I(Y;\, S \mid d)}_{\mathrm{CI}(d)\text{: boundary increment}} + \underbrace{I(Y;\, h \mid d,\, S)}_{\text{hierarchical increment}}
\end{aligned}$$

The **hierarchical increment** $I(Y; h \mid d, S)$ is the additional uncertainty reduction from knowing how far into the heading structure a boundary crossing reaches, beyond what is provided by knowing the continuous distance and whether any boundary was crossed at all. It measures the cognitive value of polyhierarchical organization as a quantity over and above gross categorical partitioning.

Implication for H4. The hierarchical increment formalizes the theoretical claim underlying H4: `hier_distance` carries cognitive information about sense identity that is not redundant with either continuous semantic distance or the binary heading congruence indicator. A positive and significant hierarchical increment means that LCSH's polyhierarchical structure encodes cognitively real distinctions above and beyond its gross categorical partitioning. A zero increment would indicate that knowing the depth of the boundary crossing adds nothing once the binary fact of crossing is known—a finding with direct policy implications for whether the engineering overhead of maintaining a full polyhierarchical structure is justified by its cognitive contribution to user disambiguation.

Implication for H5. The hierarchical increment also provides a formal account of ambiguity-type moderation. Homonymous pairs, which typically map to headings in entirely separate hierarchical subtrees ($h = \infty$ by convention), produce a categorical structure that maximizes $I(Y; h \mid d, S)$ relative to polysemous pairs, which may cross a boundary while remaining hierarchically proximate. The predicted ambiguity-type difference in boundary effects is therefore a prediction about the distribution of hierarchical increment magnitudes across ambiguity types—a restatement of H5 in information-theoretic terms that makes its structural basis explicit and distinguishes it from any claim that homonymy and polysemy are inherently different cognitive categories.

3.5.4. Pointwise Cognitive Information and the Heading Reform Diagnostic

The extensions above operate at the level of expectations and averages. A complementary diagnostic operates at the level of individual observations. For a specific stimulus pair with observed values (d_i, S_i, Y_i), define the **pointwise cognitive information**:

$$\mathrm{pCI}_i = \log \frac{P(Y_i \mid d_i,\, S_i)}{P(Y_i \mid d_i)}$$

This is the pointwise mutual information between the heading congruence and the sense identity for item i, conditional on the observed distance. It quantifies how much the specific LCSH heading assignment for this pair shifted the posterior probability toward the correct sense identification, relative to the baseline established by continuous similarity alone. Three interpretive regimes are distinct:

- If $\mathrm{pCI}_i > 0$: The heading assignment was consistent with the behavioral response — the LCSH categorization correctly concentrated posterior probability on the observed outcome.
- If $\mathrm{pCI}_i \approx 0$: The heading assignment was uninformative — categorical status neither helped nor harmed prediction beyond what distance already captured.
- If $\mathrm{pCI}_i < 0$: The heading assignment was inconsistent with the behavioral response — the LCSH categorization shifted posterior probability away from the observed outcome.

pCI functions as a "surgeon's scalpel" in assessment and KOS reform while $\mathrm{CI}(d)$ is the system-level score: $\mathrm{CI}(d)$ tells us whether LCSH is working in the aggregate; pCI tells us which specific headings are responsible for failures when it is not, or for successes when it is.

The expected value of pCI_i over all items at distance d equals $\mathrm{CI}(d)$, so pointwise and aggregate measures are formally consistent. The diagnostic value of pCI is its per-item resolution: items with systematically negative pCI are empirically identified candidates for heading revision—not on grounds of literary underrepresentation or terminological inconsistency, but because their current categorical assignment conflicts with how users process the underlying sense distinction.

> **Abridged:** A ranked list of LCSH heading assignments by pCI provides vocabulary editors with an empirically prioritized revision agenda—a quantitative operationalization of cognitive warrant as an editorial criterion, alongside the literary and cultural warrant considerations that currently guide LCSH policy.

The per-item resolution of pCI opens a further application beyond diagnosis. The fitted mixed-effects model that produces pCI scores serves a dual function: it is both a diagnostic and a surrogate for predicting the cognitive-information consequences of proposed graph edits to the LCSH DAG, without requiring new behavioral data collection after each proposed change. This surrogate capacity means the model can in principle be extended into a prescriptive optimization procedure rather than a purely descriptive one. The result is a ranked agenda specifying not only *which* heading assignments are cognitively problematic, but *what* editorial action—reassignment, record splitting, or BT/NT restructuring—would most improve the aggregate cognitive warrant score. This *surrogate-assisted pCI-gradient ascent procedure*, and the combinatorial optimization framework it requires, are developed in a companion technical document and are outside the scope of the present study; once run, the behavioral experiments described here will provide the fitted model that makes the procedure executable.

3.5.5. Pointwise Cognitive Information: Behavioral Foundations and Pre-Experimental Distinctions

The pCI formula involves three variables whose origins must be kept clearly distinct. Y_i comes from Trott and Bergen's experimental design—the ground-truth sense identity (the `same` variable)—known before any analysis. d_i comes from their BERT processing pipeline—the distributional semantic distance (`distance_bert`) between contexts, also pre-experimental. S_i comes from the LCSH mapping work—the heading congruence assigned through professional cataloging judgment and validated through librarian consultation, also pre-experimental in the sense of being determined before any participant responds.

But the probabilities in the pCI ratio—$P(Y_i \mid d_i)$ and $P(Y_i \mid d_i, S_i)$—are both estimated from behavioral data. They are item-level fitted values from mixed-effects models trained on participant response times. The pre-experimental data can validate the structural plausibility of the inputs, but only behavioral data can calibrate how those inputs translate into cognitive processing probabilities. pCI is therefore not a *structural* measure of LCSH's organization. It is a *behavioral* measure of whether LCSH's categorical congruence structure tracks how human cognitive systems process sense distinctions—the operationalization of cognitive warrant in its most precise form.

3.5.6. The Redundancy Bound

By definition, $\mathrm{CI}(d) = I(Y;\, S \mid d)$. Applying the symmetry property of conditional mutual information (see Section 3.2.4):

$$I(Y;\, S \mid d) = H(S \mid d) - H(S \mid Y,\, d) \tag{3.4}$$

Since conditional entropy is non-negative for any random variables, $H(S \mid Y,\, d) \geq 0$. Substituting into Equation 3.4:

$$\mathrm{CI}(d) = H(S \mid d) - H(S \mid Y,\, d) \leq H(S \mid d) \tag{3.5}$$

Equality holds if and only if $H(S \mid Y,\, d) = 0$, which occurs if and only if S is a deterministic function of the pair (Y, d).

$H(S \mid d)$ is the residual entropy of heading congruence after conditioning on semantic distance — the portion of S not already determined by d. The **redundancy bound** (Equation 3.5) establishes that $\mathrm{CI}(d)$ cannot exceed this residual: the information heading congruence provides about Y is bounded above by the information heading congruence provides beyond d. The name reflects the interpretive role of $H(S \mid d)$: it measures how non-redundant S is relative to d. When S is nearly collinear with d—when heading congruence is largely predictable from BERT distance alone—the residual $H(S \mid d)$ is small, and the bound suppresses $\mathrm{CI}(d)$ toward zero regardless of any genuine cognitive effect of categorical structure. This makes the empirical relationship between `same_heading` and `distance_bert` a design-validity check, not merely a descriptive statistic: if the two are strongly collinear, the study lacks the statistical headroom to detect a nonzero $\mathrm{CI}(d)$ even if one exists.

The redundancy bound thus establishes that a categorical congruence structure can carry cognitive information only to the extent that its assignments are not fully predictable from continuous similarity. Cognitive warrant therefore requires *informational non-redundancy*: the heading congruence must encode structure about sense identity that is unavailable in, or orthogonal to, continuous distributional similarity.

This is not a requirement that LCSH structure must be *uncorrelated* with BERT distance to have cognitive value. Moderate correlation is fully compatible with positive CI. What the bound rules out is the limiting case where the heading congruence is entirely predictable from distance, which would make LCSH a redundant overlay on distributional semantics rather than an independent source of information.

3.6. Cognitive Warrant and the Problem of Nomic Measurement

A recurring challenge in any measurement program is the problem of *nomic measurement*, which was described by scholar of history and philosophy of science Hasok Chang in the context of early thermometry (2004). The problem arises whenever a researcher wishes to measure a quantity X that is not directly observable and must be inferred from an observable proxy Y through some functional law $X = f(Y)$—but the form of f cannot be tested without prior knowledge of X, the very quantity that measurement is supposed to establish. The circularity is fundamental: validating the instrument requires knowing what the instrument measures, but knowing what the instrument measures requires a validated instrument.

Trott and Parkinson-Coombs (2026) identify an analogous problem in the construction of benchmarks for evaluating large language model capabilities. A benchmark score is an observable quantity; the underlying capability it purports to reflect is not. As they demonstrate, the functional mapping between the two—whether it is linear, logarithmic, sigmoidal, or categorical—is rarely specified in advance and cannot be recovered from the benchmark data alone. They argue that progress on this problem requires a process of *epistemic iteration* (Chang, 2004): rather than seeking a single axiomatic justification for the scale, researchers make progress by iteratively refining instruments, theoretical commitments, and external validation criteria in dialogue, each stage provisional but functional enough to support the next.

The cognitive warrant framework developed in this paper confronts the same problem in a different domain. The construct of interest—the degree to which a knowledge organization system's categorical congruence structure corresponds to psychologically real discontinuities in semantic processing—is not directly observable. What is observable is a set of behavioral responses (response times, appropriateness ratings) and a set of structural measurements (BERT cosine distance, LCSH heading congruence, hierarchical distance). The question is: what functional mapping between these observables and the underlying construct does the framework license?

The information-theoretic restatement of cognitive warrant provides a partial answer. Defining $\mathrm{CI}(d) = I(Y; S \mid d)$—the conditional mutual information between sense identity Y and the heading congruence indicator S given continuous semantic distance d—specifies the functional form from theoretical commitments rather than from assumption or convention. The mapping is not asserted; it is derived from the properties of conditional entropy. Three features of this specification are directly relevant to the problem Trott and Parkinson-Coombs identify.

First, $\mathrm{CI}(d)$ is a *graduated* scale in precisely the sense Chang's analysis requires. Warrant efficiency $\eta(d) = \mathrm{CI}(d) \ / \ H(Y \mid d)$ (see Section 3.5.1) maps the categorical contribution of any boundary system to a normalized interval $[0, 1]$: a score of $\eta = 0$ indicates that the heading congruence is entirely redundant with continuous distributional similarity, while $\eta = 1$ indicates that the heading assignment perfectly resolves all residual disambiguation uncertainty. This is not an ordinal ranking of boundary systems; it is a ratio-scale quantity with a principled zero and a principled maximum, both derived from the information-theoretic formalism.

Second, the functional form is *theoretically grounded* rather than arbitrarily assumed. Trott and Parkinson-Coombs note that the core epistemological challenge in graduating a benchmark is specifying what licenses the inference from a measured interval to a true interval. For $\mathrm{CI}(d)$, this license comes from conditional mutual information's formal properties: non-negativity, the chain rule decomposition, and binary entropy concavity jointly entail the expected shape of the cognitive warrant function across the semantic distance continuum. Hypothesis 3 in this study—that boundary effects are largest at intermediate distances—is not an empirical conjecture layered on top of the framework; it is a mathematical consequence of these properties (see Section 3.2).

Third, the framework satisfies the *minimal empirical criteria* that Trott and Parkinson-Coombs identify as the productive path forward in thermometry: rather than requiring a single ground-truth anchor, the cognitive warrant construct admits multiple independent sources of validation. Experiment 1 (implicit processing difficulty) and Experiment 2 (explicit appropriateness rating) operationalize the same underlying quantity through orthogonal behavioral channels. Convergence across them (H7) is the cross-instrument comparability check that epistemic iteration relies on. Trott and Bergen's (2023) Supplementary Analysis 3—showing that categorical sense information carries conditional mutual information irreducible to surprisal, a continuous probabilistic measure—constitutes prior empirical support for the same theoretical structure from an independent measurement approach.

The analogy is instructive but bounded. Temperature is a well-defined physical quantity with stable fixed points; cognitive warrant, like LLM capability, is a theoretical construct whose objective grounding is part of what the research program must establish. This study does not claim to have solved the graduation problem for KOS evaluation. It claims to have made a move in the epistemic iteration that solving it will require: specifying a functional form from explicit theoretical commitments, identifying multiple independent validation instruments, and preregistering the inferential criteria that would constitute progress. That the problem remains open is not a limitation unique to this framework—it is the condition under which measurement science in any new domain must begin.

3.7. Hypotheses

The hypotheses developed from the cognitive warrant framework evaluate whether the categorical congruence structure imposed by LCSH aligns with continuous semantic similarity and human cognitive responses. The hypotheses are organized into four tiers: representational structure, continuous–categorical interaction, structural refinements, and cross-measure convergence. The two experiments address partially overlapping hypothesis sets, but Experiment 2's set is deferred to after its design is explained in Section 4.4.

3.7.1. Representational Structure

> **H1.** *Continuous similarity effect*
> Greater BERT-derived semantic distance between paired stimuli will be positively associated with processing difficulty (`log_rt`).

H1 states the continuous component of hybrid meaning theory: distributional semantic distance predicts cognitive responses because greater semantic divergence between senses increases the effort required to integrate contextual meaning.

> **H2.** *Categorical boundary effect*
> Stimulus pairs assigned to different LCSH subject headings (`same_heading` $= 0$) will be associated with greater processing difficulty than pairs assigned to the same heading (`same_heading` $= 1$), controlling for `distance_bert`.

H2 tests whether categorical boundaries imposed by LCSH correspond to psychologically meaningful sense distinctions. In Trott and Bergen's design, categorical boundaries corresponded to distinct dictionary headword entries; here, the proposed boundary signal derives from LC subject heading assignments produced through professional cataloging practice.

3.7.2. Continuous–Categorical Interaction

> **H3.** *Nonlinear boundary–distance interaction*
> The magnitude of the LCSH categorical boundary effect will vary nonlinearly as a function of `distance_bert`, such that boundary-related processing costs are strongest at intermediate semantic distances (provisionally 0.3-0.6 cosine distance) and attenuated at both low and high distances.

H3 is not an empirical prediction layered on top of the CI framework—it is a *mathematical consequence* of it. The argument proceeds in three steps under a single minimal assumption: that the probability of a same-sense judgment decreases monotonically and smoothly with semantic distance, formalized as $P(Y = 1 \mid d) = \sigma(\alpha - \beta d)$ for $\alpha, \beta > 0$, where $\sigma(\cdot)$ is the logistic function. This encodes nothing more than the claim that BERT cosine distance is directionally valid as a measure of semantic dissimilarity—a commitment the study's measurement framework requires regardless of H3.

Under this assumption: (1) $H(Y \mid d)$, the disambiguation uncertainty at distance d, is unimodal—zero when sense identity is already certain at the distance extremes, and positive in the interior where contexts are genuinely ambiguous; (2) $\mathrm{CI}(d) = I(Y; S \mid d)$ is squeezed between zero and $H(Y \mid d)$ by the non-negativity of conditional mutual information, so it too must be zero at both extremes; and (3) a continuous function that is zero at both endpoints and positive somewhere in the interior is forced by continuity to achieve its maximum at some interior point. The inverted-U shape is therefore the *only* shape consistent with these three properties. No functional form is imposed; it is derived.

This has a specific consequence for how H3 should be interpreted against Trott and Bergen's (2023) null result on the analogous interaction. Their transformation analyses compared eight functional forms for how sense boundaries might warp the distance–RT relationship, ranging from additive step-shifts and multiplicative rescaling to a linear interaction term ($D \times SB$). All of these, however, share a common property: they apply a monotone modulation of the boundary's contribution across the distance continuum. The additive transformation displaces same- and different-sense distances by the same constant regardless of distance value. The multiplicative transformation scales by a fixed factor. The linear interaction changes the slope by a constant amount. None of these functional forms is capable of producing an effect that peaks at intermediate distances and falls to zero at both extremes. That non-monotone, inverted-U pattern is precisely what H3 predicts, and it is untested by any of Trott and Bergen's eight functional forms. Their null result on $D \times SB$ does not compete with H3; together, their full set of transformation analyses leaves H3 untested.

More importantly, the regression operationalization of H3 is not asking whether the peak *exists*—the mathematics establishes that it must, if H2 holds at all. The regression is asking whether the peak is large enough to detect at the study's sample size, and where in the distance continuum it falls. A significant nonlinear form confirms not merely an empirical pattern but the mathematical structure that entailed it. A null result, conversely, localizes the misfit: either the logistic monotonicity assumption breaks down in this stimulus domain, or $\mathrm{CI}(d) \approx 0$ everywhere—which would be the joint failure of H2 and H3, not an anomaly specific to the nonlinear term. H2 and H3 are logically linked in exactly this way: confirming H2 (that the boundary adds any predictive value at all beyond continuous distance) commits the framework to H3's shape as a theorem rather than a conjecture.

3.7.3. Structural Refinements

> **H4.** *Within-category hierarchical distance effect*
> Among stimulus pairs whose LCSH headings share a BT ancestor within three hierarchical levels (`same_nbhd` = 1), greater `hier_distance` will be positively associated with processing difficulty (`log_rt`), independent of `distance_bert`.

H4 is the structural analog of Trott and Bergen's (2023) sense attraction test. In their exploratory analysis of same-sense pairs, they found that `distance_bert` continued to predict RT, establishing that within-category distributional distance carries genuine behavioral signal. H4 asks the corresponding question for LCSH hierarchical structure: does `hier_distance` carry behavioral signal *within* the same-neighborhood group, above and beyond what `distance_bert` already captures? If it does, the LCSH polyhierarchy encodes conceptual proximity information that is (a) not reducible to distributional similarity and (b) perceptible to users as a gradient of categorical fit, not merely as a present-or-absent boundary. A confirmed H4 alongside Frame 2's predicted reduction in within-category BERT distances would provide converging evidence that LCSH categorical labels modulate representational space in the manner the sense attraction account predicts—and that the modulation operates through hierarchical position, not just boundary membership.

> **H5.** *Ambiguity type moderation of boundary effects*
> The magnitude of categorical boundary effects will be stronger for homonymous stimulus pairs than for polysemous pairs, reflecting the larger hierarchical distances typically associated with homonymous boundary-crossing in LCSH.

Trott and Bergen (2023) found no significant interaction between sense boundary status and ambiguity type: their well-powered design produced a null result on the question of whether homonyms and polysemes generate different processing effects at the boundary. H5 predicts a moderation by ambiguity type in the present study—but not because the present design proposes a different cognitive architecture. The prediction is *structural*, and the distinction matters.

This structural prediction derives from the organizational properties of the measurement instrument rather than from a claim about the cognitive system's internal representations. In this study, the measurement instrument is LCSH's polyhierarchical DAG. When two word senses are homonymous—sharing a form but no semantic ancestry—their LCSH authority records are typically located in entirely separate hierarchical subtrees with no recoverable common ancestor at any finite depth (`hier_distance` $= \infty$ or very large). This is not arbitrary: the lexicographic grounds for classifying two senses as homonymous are precisely that they lack the kind of semantic relatedness that would place them in the same region of a subject vocabulary. LCSH's polyhierarchical structure encodes this independence by assigning the senses to branches that do not converge.

Polysemous boundary-crossings have a different structural signature. Polysemous senses are related meanings derived from a common root; their LCSH headings are therefore often hierarchically proximate even when they are formally assigned to different authority records. A word like *bank* in its financial and geographical senses (*Banks and banking* vs. *Stream banks*) is homonymous and maps to branches with no shared hierarchical ancestor. A word like *interest* in its financial and psychological senses may cross a categorical boundary while both headings remain within a broader region of the subject vocabulary—polysemous senses may be formally distinct but structurally adjacent.

H5 therefore does not claim that the cognitive system distinguishes homonymy from polysemy directly. It claims that homonymous boundary-crossings are systematically associated with larger `hier_distance` values than polysemous ones, and that—if H4 holds—larger hierarchical distance predicts greater processing difficulty. The ambiguity type moderation is predicted to emerge as a consequence of this distributional asymmetry in LCSH structure, not as an independent cognitive effect of sense type.

This framing places H5 in a specific relationship to Trott and Bergen's null result. Their design used dictionary headword entries as categorical boundaries, a representation that does not encode hierarchical distance: a separate-entry pair (homonymous) and a same-entry pair (polysemous) differ categorically but not structurally in a dictionary. The null result is therefore entirely consistent with a structural account: without the hierarchical amplification that LCSH introduces, there is no mechanism by which ambiguity type should modulate boundary effects, and none was found. H5's predicted moderation is not a challenge to their finding—it is a consequence of replacing a flat categorical instrument with a hierarchically organized one.

> **H6.** *Hierarchical distance as mediator of ambiguity type moderation*
> If stronger boundary effects for homonymous items are driven by the larger hierarchical distances associated with those crossings, the interaction between ambiguity type and `same_heading` should diminish or disappear once `hier_distance` is included as a covariate.

H6 is the critical test that distinguishes the structural account from a cognitive one. The structural account holds that ambiguity type has no direct effect on processing: it operates entirely through `hier_distance`, because homonymous pairs happen to have larger hierarchical distances in LCSH, and larger hierarchical distances happen to predict greater processing difficulty. On this account, entering `hier_distance` as a covariate should absorb the ambiguity type moderation entirely—the `same_heading` × `ambiguity_type` interaction should attenuate substantially or become non-significant once `hier_distance` is controlled.

The cognitive account holds that ambiguity type does something additional: that the cognitive system represents homonymy and polysemy differently at the categorical boundary in a way that is not reducible to the structural position of the headings. On this account, the interaction should survive the hierarchical distance control.

The theoretical stakes of these outcomes differ sharply. A mediated result—both H5 and H6 confirmed—is the *more parsimonious* outcome, requiring no new cognitive mechanism beyond what H1–H4 already establish. It would simultaneously explain Trott and Bergen's null result (their design lacked the hierarchical variable that drives the effect) and provide a principled account of why dictionary-based and LCSH-based boundary tests yield different ambiguity type profiles. A non-mediated result—H5 confirmed, H6 disconfirmed—would be genuinely surprising: it would imply a cognitive distinction between homonymy and polysemy that Trott and Bergen's well-powered study failed to detect, and would require stronger theoretical justification than the current framework provides. The structural account is accordingly the more conservative prediction; H6 is designed to test whether the data require anything more.

3.7.4. Cross-Experiment Convergence

> **H7.** *Convergence across implicit and explicit measures*
> Items for which crossing an LCSH categorical boundary produces greater implicit processing costs in Experiment 1 will also yield larger discrimination scores in Experiment 2. Operationalized as a positive item-level Pearson correlation between the implicit disruption index ($\overline{\mathrm{RT}}_{\text{diff-sense},k} - \overline{\mathrm{RT}}_{\text{same-sense},k}$) per item k and Δr_k.

H7 is the construct validity capstone of the entire study. If the same stimulus items that produce large RT boundary costs in Experiment 1 also produce large discrimination scores in Experiment 2—across entirely different task demands, response modalities, and dependent variables—then cognitive warrant cannot be explained as a task artifact. Item-level correlation is the right operationalization here because it is the only test that holds constant the properties of specific word senses while varying the measurement approach, providing multi-method triangulation of CI(d) as a stable construct rather than a paradigm-specific measurement artifact. The Experiment 2 side of this correlation—the discrimination score Δr_k and the foil structure that generates it—is defined in Section 4.4, where the constructs presupposed by E2-H1 through E2-H5 are also introduced; those hypotheses are stated there rather than here for the same reason.

3.7.5. Connections Among the Hypotheses

The research design tests Trott and Bergen's hybrid meaning theory in a KOS context while adding LCSH-specific elements:

1. **Continuous effects** (H1, E2-H3): BERT distance predicts cognitive responses. Replicates the continuous component of the hybrid account.
2. **Categorical effects** (H2, E2-H1): LCSH boundaries create processing discontinuities independent of semantic distance. Replicates the categorical component with professionally imposed rather than lexicographic boundaries.
3. **Interaction effects** (H3): Boundary effects peak at intermediate distance. Tests the disambiguation-load hypothesis requiring a nonlinear specification that Trott and Bergen's linear model could not detect.
4. **Structural moderation** (H4–H6, E2-H2, E2-H4, E2-H5, E2-H6): Hierarchical distance and ambiguity type refine boundary effects. Exploits LCSH-specific structure absent from dictionary-based designs.
5. **Convergence** (H7): RT effects and rating effects converge at the item level. Provides multi-method triangulation of cognitive warrant as a construct.

3.8. Chapter Summary

By situating LCSH within a continuous–categorical representational framework and formalizing cognitive warrant as uncertainty reduction, this chapter generates the study's empirical logic. The research questions are logically necessary tests of the integrated model. The representational tier requires independent effects of graded semantic distance and categorical boundary status (RQ1). The interactional tier requires a quadratic boundary–distance interaction (RQ2). The structural tier requires that hierarchical distance and ambiguity type systematically refine boundary effects (RQ3). The convergence tier requires that implicit and explicit measures align at the item level (RQ4). Each tier corresponds to a specific set of hypotheses (H1–H7, E2-H1–E2-H5) formalizing these requirements as directional predictions across both experiments. Chapter 4 describes how each hypothesis is operationalized in the experimental design and analytical plan.

CHAPTER 4: METHODS

This chapter describes the methodology for empirically testing whether LCSH categorical boundaries align with users' cognitive processing of polysemes and homonyms. The study employs a five-phase approach: (1) LCSH mapping of stimulus word senses and expert validation, (2) LCSH hierarchical graph construction and structural variable computation, (3) two complementary behavioral experiments adapting Trott and Bergen's (2023) research design to operationalize cognitive warrant through implicit processing difficulty and explicit categorical alignment, (4) BERT embedding of stimulus contexts and LC subject headings to compute semantic distances, and (5) data analysis using hierarchical linear modeling.

4.1. LCSH Mapping Procedure

4.1.1. Candidate Retrieval

For each of the 208 word senses drawn from Trott and Bergen's materials, candidate LCSH headings are retrieved via the LC Linked Data Service `suggest2` API (id.loc.gov). For each candidate, the scope note, BT/NT relations, and RT links are inspected. RT links are used qualitatively during heading selection—a candidate whose RT links reference concepts consistent with the target sense is preferred—but excluded from the quantitative distance graph.

4.1.2. Granularity Differences Between LCSH and the Source Dictionaries

Translating Trott and Bergen's (2023) dictionary-derived sense boundaries into LCSH categorical boundaries requires confronting a fundamental structural asymmetry between the two systems. Dictionaries and LCSH are organized according to different representational logics and serve different epistemic purposes, with the consequence that their sense boundaries do not stand in a simple correspondence relation. Understanding the nature and scope of this asymmetry is necessary both for making principled mapping decisions during stimulus development and for correctly interpreting cases where the two systems diverge.

The fundamental question each system answers is different — "what does this word mean?" versus "what is this document about?" — and their sense boundaries reflect those divergent purposes. Rather than treating the mismatches as methodological problems to be minimized, however, the study treats them as structurally informative: they constitute natural variation in the degree to which a professional controlled vocabulary tracks the sense distinctions that lexicographers have independently identified, and that variation is itself empirically tractable through the response time and appropriateness rating data the study collects.

4.1.3. Decision Criteria: Six-Criterion Scoring Matrix

When multiple candidate headings are retrieved for a given word sense, each candidate is evaluated using a structured scoring matrix with six dimensions (each scored 0–3, with semantic fit as the primary decision criterion in near-tie cases):

1. **Semantic fit**: Does the heading match the specific sense established by context sentences?
2. **USE/UF variant coverage**: Does the heading's reference structure capture lexical variants?
3. **Hierarchical coherence**: Are BT/NT relations consistent with the meaning intended?

4. **Specificity**: Is the heading appropriately specific (not overly broad or narrow)?
5. **Scope note clarity**: Does any scope note confirm or restrict the application?
6. **Boundary correspondence**: If two senses require different headings, do the headings clearly differentiate them?

4.1.4. Expert Validation and Inter-Rater Reliability

Two academic librarians with LCSH expertise independently map a 20% random sample of the full stimulus set. Agreement is assessed using Cohen's κ. If $\kappa < 0.80$, mapping guidelines are refined and problematic items are remapped until the threshold is reached. Discrepancies are resolved through structured consensus discussion.

4.2. Construction of the LCSH Hierarchical Graph

4.2.1. Characterization of the BT/NT Subgraph as a DAG

LCSH exhibits polyhierarchy: a single authority record may carry more than one Broader Term entry. For the present study's analytical purposes, the BT/NT subgraph is most accurately characterized as a *directed acyclic graph (DAG)*: directed because BT/NT is asymmetric, acyclic because genuine hierarchical cycles are cataloging errors, and unconstrained with respect to in-degree, accommodating polyhierarchy without structural modification

The DAG representation supports the two algorithms central to this study: the Lowest Common Ancestor (LCA) algorithm (identifying the most specific shared ancestor) and shortest-path length computation. Both are well-defined on DAGs and are implemented in NetworkX's `DiGraph` class.

> **Abridged:** Young (2017) identifies several structural shortcomings of LCSH including the unexamined programmatic generation of BT/NT links in the mid 1980s. The practical implication is that `hier_distance` is a noisier proxy for genuine conceptual proximity than it would be if derived from a purpose-built thesaurus (e.g., MeSH) whose BT/NT links were established through systematic facet analysis. The key is that the flaws are not invalidating but require documentation in analyses and consideration when interpreting results of cognitive warrant metrics.

4.2.2. Hierarchical Distance Computation

`hier_distance` is defined as the sum of shortest path lengths from each of two LCSH authority records to their LCA in the BT/NT subgraph:

$$\texttt{hier\_distance}(i, j) = d_{\mathrm{BT}}(i, \mathrm{LCA}(i, j)) + d_{\mathrm{BT}}(j, \mathrm{LCA}(i, j))$$

where d_{BT} is the number of BT hops from a heading to the LCA. Example values: `hier_distance` $= 0$ (same heading); $= 1$ (direct BT/NT relationship); $= 2$ (siblings sharing an immediate BT); $= \infty$ (no common ancestor recoverable in the graph, typical for homonymous pairs in disconnected subtrees). Pairs with `hier_distance` $= \infty$ receive `same_nbhd` $= 0$ and are excluded from hierarchical-distance analyses while serving as the cross-category contrast group. Table 4.1 provides example LCSH mappings of stimulus words with their semantic and hierarchical distances.

4.2.3. The `same_heading` and `same_nbhd` Variables

Motivation: the within-category gradient problem

Trott and Bergen's (2023) exploratory analyses tested whether `distance_bert` predicts response time *within* same-sense pairs—that is, holding `same` status constant

Table 4.1: Example LCSH mappings for eight stimulus words, illustrating the range of sense relationships in the stimulus set. Each row shows one specific prime/target sentence pair (of up to four cross-sense combinations available per word); d is the BERT cosine distance between exactly these two sentence contexts, not a word-level average. Authorized headings are exactly as assigned in the LCSH mapping procedure. h is hierarchical distance (BT/NT shortest path) between the two headings, with $h = \infty$ indicating no shared broader-term ancestor within the search horizon.

Word	Ambiguity	Example sentence pair	Authorized LCSH heading	d	h
bat	Homonymy	*He saw a wooden bat.* *He saw a furry bat.*	Baseball bats Bats	0.141	∞
company	Polysemy	*He liked the investment company.* *He liked the pleasant company.*	Business enterprises Fellowship	0.489	8
course	Polysemy	*It was an introductory course.* *It was a collision course.*	Education Navigation	0.477	∞
drill	Homonymy	*It was a grueling drill.* *It was an electric drill.*	Drill and minor tactics Electric drills	0.255	∞
lamb	Polysemy	*They liked the cute lamb.* *They liked the marinated lamb.*	Lambs Lamb (Meat)	0.091	1
orange	Polysemy	*It was a juicy orange.* *It was a bright orange.*	Oranges Orange (Color)	0.307	∞
pick	Polysemy	*He picked his scab.* *He picked his coffee.*	Compulsive skin picking Decision making	0.160	8
port	Homonymy	*It was a delicious port.* *It was a windy port.*	Port wine Harbors	0.225	11

and asking whether continuous distance still matters. They found that it does: even within sense categories, closer contexts of use yield faster responses. This result favors the sense attraction account over sense distillation, establishing that within-category variance in distributional distance carries genuine behavioral signal.

The present study seeks an analogous test using LCSH structure: does *hierarchical distance* predict RT within "same-category" pairs? This would establish a "within-category" gradient in LCSH-defined conceptual space analogous to the continuous gradient Trott and Bergen found in distributional space.

Using `same_heading` as the grouping variable for this analysis, however, fails for three independent reasons. First, pairs coded `same_heading` = 1 (same authority record) have `hier_distance` = 0 by definition—the two senses map to a single DAG node, so there is no path to compute. Restricting the analysis to `same_heading` = 1 pairs collapses the within-category distribution to a single degenerate value; there is no gradient to estimate a slope over. Second, many homonymous pairs coded `same_heading` = 0 (different authority records) map to entirely disconnected subtrees of the LCSH DAG, yielding `hier_distance` = ∞; these must be excluded from any hierarchical-distance model, but `same_heading` provides no principled criterion for doing so. Third, the scientific value of H4 lies precisely in testing whether *degree* of hierarchical separation—not merely its presence or absence—predicts behavioral responses, which requires a "within-category" sample in which `hier_distance` varies continuously over a meaningful range.

Definition of `same_heading`

`same_heading` (binary): = 1 if both senses in a pair map to the same LCSH authority record; = 0 if they map to different authority records. This directly mirrors

Trott and Bergen's (2023) `same` variable, with LCSH heading assignments substituted for dictionary headword entries. `same_heading` is the boundary indicator for H1, H2, and H3, where the research question concerns whether stimuli were assigned to formally distinct indexing units.

Definition of `same_nbhd`

`same_nbhd` (binary): $= 1$ if both senses in a pair map to the same LCSH authority record *or* to authority records that share at least one broader term (BT) within $D = 3$ hierarchical levels; $= 0$ otherwise. Under this definition, pairs coded 1 constitute a semantically proximate "neighborhood" in which `hier_distance` is well-defined and varies from 0 (same record) up to 6 (two paths of three BT steps each to the lowest common ancestor); pairs coded 0 fall outside that region, serving as the cross-category contrast group in H5 and H6 and as the basis for E2-H2.

When the shortest-path algorithm finds no common BT connecting two authority records within D levels—the expected outcome for many homonymous pairs, such as the financial and river senses of the word *bank*—those pairs receive `same_nbhd` $= 0$ regardless of D. Genuinely disconnected subtrees constitute the clearest instance of cross-category membership; their assignment to the contrast group is theoretically informative rather than a data loss.

The two variables are related but non-equivalent. Every pair with `same_heading` $= 1$ is also `same_nbhd` $= 1$, because identity of authority record entails hierarchical proximity. The converse does not hold: pairs of *distinct* authority records sharing a BT within three levels are coded `same_heading` $= 0$ (different records) but `same_nbhd` $= 1$ (within the categorical neighborhood). This asymmetry is intentional. `same_heading` preserves strict correspondence with Trott and Bergen's operationalization wherever the research question concerns formal boundary status; `same_nbhd` extends it to accommodate LCSH's hierarchical gradient for the within-category analyses that H4, H5, H6, E2-H4, and E2-H5 require.

Selection of $D = 3$

The threshold D determines how large the hierarchical neighborhood is. Setting D too low starves the within-category sample of observations and compresses the `hier_distance` distribution, reducing power for H4; setting it too high admits pairs whose lowest common ancestor is a vacuously broad superordinate—terms such as *Knowledge* or *Natural sciences and mathematics*—such that co-membership in the within-category group reflects shared broad classification rather than genuine semantic proximity. Four considerations motivate $D = 3$ as the default.

1. **Structural fit**: Topical LCSH headings typically reside three to six levels below a disciplinary root (e.g., *Psychology* $\rightarrow$ *Cognitive psychology* $\rightarrow$ *Memory* $\rightarrow$ *Short-term memory*). At $D = 3$, the shared ancestor typically names a recognizable subject area rather than an uninformative superordinate—the "second-cousin" proximity at which professional cataloging judgment treats headings as conceptually related.
2. **Gradient preservation**: $D = 3$ generates `hier_distance` values spanning $\{0, 1, 2, 3, 4, 5, 6\}$, providing sufficient range for the mixed-effects slope estimate in H4. Values of $D < 2$ would likely compress this distribution into $\{0, 1, 2\}$ for the proof-of-concept subset, substantially reducing statistical power.
3. **Protection against vacuous ancestors**: $D \geq 4$ risks including pairs whose LCA carries no semantic specificity, making co-membership in the within-category group uninformative about genuine conceptual proximity.

4. **Empirical calibration**: After graph construction, the observed `hier_distance` distribution will be inspected for natural breaks. If the primary cluster terminates at $D = 2$ or $D = 4$, the threshold will be adjusted and reported as a preregistered deviation; $D = 3$ is retained as a sensitivity condition in all cases.

4.3. Experiment 1: Primed Sensibility Judgment Task

4.3.1. Purpose

Experiment 1 is meant to fully replicate Trott and Bergen's (2023) experiments except for one factor: LCSH is used to categorize word uses instead of dictionaries. From a participant's perspective, the experiments are indistinguishable.

4.3.2. Dependent and Independent Variables

Primary DV: Log-transformed RT (`log_rt`) on correct target responses. *Primary predictors*: `distance_bert` (Frame 1 continuous distance), `same_heading` (binary boundary), `hier_distance` (continuous, within-`same_nbhd` = 1 pairs), `same_nbhd` (binary neighborhood), `ambiguity_type`. *Covariate*: `prior_rt` (mean RT across participants to the same sentence when presented as prime).

4.4. Experiment 2: Heading Appropriateness Rating Task

4.4.1. Purpose

Experiment 2 operationalizes cognitive warrant through explicit categorical judgment. Participants directly assess whether LCSH headings are appropriate for organizing content matching a specific word sense, providing a conscious measure of cataloger–user alignment.

Experiment 2 introduces a novel **appropriateness rating paradigm** that directly measures cataloger-user alignment. After reading context sentences, participants rate (on a 1-7 scale) how appropriate different LCSH headings would be for indexing documents with that context. This paradigm simulates the cataloging judgment that determines how resources appear in discovery systems—providing controlled measurement of whether LCSH categorical boundaries align with users' intuitive semantic processing. By testing whether users' appropriateness judgments show categorical boundary effects, the paradigm reveals cognitive alignment between professional vocabulary structure and user cognition.

Appropriateness rating (`rating`), a raw outcome variable in Experiment 2, is a judgment by a participant of the appropriateness of a given subject heading as describing the 'aboutness' (i.e., the shared topic) of the presented stimulus sentences. This scale, however, conflates two empirically distinct quantities:

- Absolute heading fit: the degree to which the best available LCSH heading captures the target word sense at all, irrespective of alternatives. For some items—particularly verb senses such as "stake" (to tether an animal or object)—the optimal LCSH mapping identified in Experiment 1 represents a reasonable but imperfect heading because LCSH does not contain a dedicated authorized heading for this sense. A participant rating this heading at 3 of 7 may be making an accurate absolute judgment rather than failing to align with the cataloger's categorization.
- Relative discrimination: the degree to which participants appropriately rank the correct heading above foil headings, regardless of the absolute level of fit. A participant who rates the correct heading at 3 and the within-category foil at 1 demonstrates sensitivity to LCSH categorical structure even if the absolute fit is poor.

- If absolute ratings are analyzed without accounting for item-level ceiling differences in maximum achievable fit, then items with genuine LCSH coverage gaps will systematically produce lower ratings for all heading types, potentially masking cataloger-user alignment effects that are present at the relative level.

Discrimination score (`delta_score`), a relative outcome variable in Experiment 2, is defined as $\Delta r_{jk} = r_{\text{target},jk} - r_{\text{foil},jk}$ where $r_{\text{target},jk}$ is participant j's appropriateness rating of the "correct" or target heading for item k and $r_{\text{foil},jk}$ is their rating of the "foil" heading for the same item in the same session. This outcome variable is used in the data analysis instead of the appropriateness rating. This controls for item-level variation in maximum achievable heading fit given current LCSH structure. The range of delta_score is $[-6, 6]$ where

- $\Delta r > 0$ indicates that the participant appropriately differentiated the target heading from the foil.
- $\Delta r \approx 0$ indicates a failure to discriminate.
- $\Delta r < 0$ indicates a preference inversion.

4.4.2. Cognitive Mechanism

The critical cognitive difference from Experiment 1 is that participants must *explicitly* recognize and evaluate the categorical distinction the heading encodes. Where Experiment 1 tests whether LCSH boundaries correspond to psychologically real processing discontinuities (implicit), Experiment 2 tests whether users consciously endorse those same boundaries as conceptually appropriate (explicit). Together, the two experiments provide complementary evidence: Experiment 1 is sensitive to any cognitively real boundary whether or not the participant can articulate it; Experiment 2 requires the user to consciously access and evaluate the categorical logic of the heading. Convergence between the two (H7) provides the strongest evidence that cognitive warrant is a stable construct rather than a task-specific artifact.

4.4.3. Simultaneous Presentation Design

A critical design decision: context sentences are presented *simultaneously* (not sequentially). The specificity of a single context sentence from Trott and Bergen's pool can be misleading. For example, "It was a tennis match" presented independently of "It was a soccer match" could bias a participant's interpretation toward the specific sport even if a feature such as putting the target word "match" in bold type was part of the instructions. Showing both sentences together directs the participant's attention to the sense shared by both contexts.

4.4.4. Stimuli and Trial Structure

Each trial presents two sentences simultaneously establishing a specific word sense, followed by four LCSH headings to rate on a 1–7 appropriateness scale:

1. **Target heading**: The authorized LCSH heading mapped to the sense in question.
2. **Within-category foil**: A different LCSH heading with `same_nbhd` = 1 (within categorical neighborhood), with `distance_bert` matched to the cross-category foil (± 0.05).
3. **Cross-category foil**: An LCSH heading with `same_nbhd` = 0 (outside categorical neighborhood), with `distance_bert` matched to the within-category foil (± 0.05).
4. **Attention check**: A clearly irrelevant heading (e.g., *Butterfly gardening* for a financial banking context).

4.4.5. Computational Semantic Measures

Embedding Model

BERT (bert-base-uncased; Devlin et al. 2019) is used for the embedding of all stimulus sentences, following Trott and Bergen (2023). Cosine distance between [CLS] token embeddings from these contexts serves as the continuous semantic distance measure (distance_bert $= 1 -$ cosine similarity).

Sentence Frame Strategy

The BERT-distance matching of within-category and cross-category foils in Experiment 2 controls for item-level variation in LCSH coverage quality and generates directly interpretable evidence for categorical alignment. It allows categorical position to be isolated from semantic distance as the driver of foil rating differences but requires an alternative basis for semantic distance calculation: using embeddings of subject headings instead of the original stimulus contexts.

Because BERT encodes meaning contextually, raw heading strings cannot be embedded as isolated tokens without collapsing the context-sensitive representations on which the model's validity depends. For different-sense/heading pairs, LC subject headings are thus embedded in the neutral frame "The topic of this resource is [heading]" before BERT encoding. This frame ("Frame 1") mirrors the implicit propositional structure of subject cataloging and generates distance_bert as the primary continuous predictor in the regression models.

An alternative frame ("Frame 2") embeds each stimulus sentence with its subject heading using the structure "Regarding [heading]: [stimulus sentence]". This prepends the subject heading as a topical anchor before the sentential context, preserving the natural linguistic variation between M1_a and M1_b (or M2_a and M2_b) that drives within-category distance while grounding the embedding in the categorical representation supplied by the authority record. The advantage of this frame is that it accommodates both congruent and incongruent heading assignments which might be used in another analysis.

4.4.6. Dependent and Independent Variables

Primary DV: Discrimination score $\Delta r = r_{\text{target}} - r_{\text{within-cat foil}}$. *Secondary DV*: Individual heading ratings (target, foils). *Predictors*: `distance_bert_target` (Frame 1 distance from context embedding to target heading embedding), `distance_bert_foil` (Frame 1 distance from context embedding to foil heading embedding), `same_heading` (target vs. foil boundary status), `hier_distance`.

4.4.7. Experiment 2 Hypotheses (E2-H1–E2-H6)

The appropriateness rating paradigm introduces a discrimination score ($\Delta r = r_{\text{target}} - r_{\text{within-cat foil}}$) as the primary DV. The foil structure is critical: within-category and cross-category foils are matched on `distance_bert` (± 0.05), allowing categorical position rather than semantic distance to drive any observed foil differences.

> **E2-H1.** *Target preference*
> The target LCSH heading will receive significantly higher appropriateness ratings than both within-category and cross-category foils, yielding positive Δr on average.

This is the basic validity check for the entire Experiment 2 paradigm: if participants cannot reliably distinguish the correct LCSH heading from incorrect alternatives, nothing downstream is interpretable. A positive mean Δr indicates that the heading-appropriateness task is doing what it is designed to do — eliciting sensitivity to the difference between a professionally sanctioned heading and a foil. It is the explicit-judgment analog of H2 in Experiment 1, testing the categorical component of cognitive warrant through a direct rather than implicit behavioral measure.

> **E2-H2.** *Cross-category foil disadvantage*
> Cross-category foils (`same_nbhd` $= 0$) will receive lower appropriateness ratings than within-category foils (`same_nbhd` $= 1$), even with matched BERT distances, indicating that categorical position drives the judgment independently of semantic distance.

This is the critical test of whether categorical position carries independent information beyond semantic similarity in the explicit rating modality. By matching cross-category and within-category foils on BERT distance, the design isolates the contribution of heading category membership: if cross-category foils receive lower ratings at matched distances, the effect must be categorical rather than distributional. This is the explicit-judgment analog of H2 from Experiment 1 — it asks whether the cognitive system treats LCSH boundaries as real categorical distinctions even when participants are consciously evaluating indexing appropriateness rather than incidentally revealing processing difficulty.

> **E2-H3.** *Foil BERT distance effect*
> `distance_bert_foil` will be negatively associated with foil appropriateness ratings, independently of categorical boundary status, replicating H1's continuous similarity effect in the explicit rating modality.

This tests whether the continuous semantic component of hybrid meaning theory replicates in the explicit rating modality, independent of categorical structure. The prediction is that foils semantically further from the target sense (larger BERT distance) are rated less appropriate even within foil category — meaning gradient similarity information shapes explicit judgments in the same direction as it shapes implicit RT in Experiment 1. Confirming this alongside E2-H2 would establish that Experiment 2 captures both the continuous and categorical components of the hybrid model.

> **E2-H4.** *Hierarchical distance and discrimination*
> Within the same-neighborhood group, greater `hier_distance` between target and within-category foil headings will be associated with larger Δr.

Within the within-category foil group — where both target and foil headings belong to the same LCSH neighborhood — this hypothesis asks whether hierarchical separation within that neighborhood still predicts discrimination performance. It tests whether fine-grained LCSH structure (not just broad category membership) is psychologically legible to users: heading pairs that are formally in the same neighborhood but structurally farther apart in the DAG should be easier to discriminate. This is the explicit-judgment complement to H4 in Experiment 1.

4.5. Statistical Analysis Plan

4.5.1. Data Processing and Quality Control

Response time exclusions (Experiment 1): Incorrect sensibility judgments excluded; RT $<$ 200 ms or $>$ 3000 ms excluded; trials with RT $>$ 3 SD above subject mean excluded; subjects with accuracy $<$ 70% on critical trials excluded.

Rating exclusions (Experiment 2): Trials with failed attention check (attention check heading rated $\geq$ 4) excluded; subjects with $>$ 25% failed attention checks excluded.

Data quality: Bot detection scores; mobile device flag; native speaker verification.

4.5.2. Trott and Bergen's Level-1 and Level-2 Models

In a strictly nested design,[4] each lower-level unit of a hierarchical linear model belongs to one and only one higher-level unit — students nested within schools, measurements nested within patients (Raudenbush & Bryk, 2002). Trott and Bergen's design departs from this in a critical way: a single observation (trial) is the lower-level unit, but *subjects and (target) words constitute two fully crossed, independent grouping factors.* Any given subject saw a subset of items, and any given item was seen by a subset of subjects, but no subject "belongs" to a specific word and no word "belongs" to a specific subject.[5] A cross-classification arises when lower-level units simultaneously share memberships in multiple (higher-level) groups (Raudenbush & Bryk, 2002). This data structure can be presented in Raudenbush and Bryk's framework (with main effects) as follows.

Level-1 (Trial-Level) Model

$$\begin{aligned}\texttt{log\_rt}_{jk} = \pi_{0jk} + \pi_{1jk}(\texttt{distance\_bert})_{jk} + \pi_{2jk}(\texttt{same})_{jk} \\ + \pi_{3jk}(\texttt{ambiguity\_type})_{jk} + \pi_4(\texttt{prior\_rt})_k + e_{jk}\end{aligned}$$

[4]In hierarchical linear modeling, *nesting* refers to a property of the data structure: lower-level units of analysis are contained within higher-level units. In the psycholinguistic literature, by contrast, *nested models* refers to a property of competing statistical specifications: one model is nested within another when it can be obtained by constraining one or more parameters of the fuller model to zero.

[5]I prefer the term *participant* to human *subject*. Also, the regular co-appearance of the random effects of *subject* and *word*, both in text and models, might cause someone to misinterpret the former in the sense of a topic, not participant; however, I use the term *subject* to be consistent with Trott and Bergen's statistical models.

where π_{0jk} is the trial intercept (expected `log_rt` for subject j on word k when all predictors are zero) and the distributional assumption is $e_{jk} \sim \mathcal{N}(0, \sigma^2)$.

Level-2 (Between-Trial / Cross-Classified) Model

The Level-1 coefficients become outcomes at Level 2. The intercept π_{0jk} is decomposed into contributions from subjects (indexed by subscript j) and words (indexed by k). The slope coefficients π_{1jk}, π_{2jk}, π_{3jk} are given random variation over subjects (but not over words — only the intercept varies randomly over words in Trott and Bergen's specification).

$$
\begin{aligned}
\pi_{0jk} &= \theta_0 + b_{0j} + c_{0k} \\
\pi_{1jk} &= \theta_1 + b_{1j} \\
\pi_{2jk} &= \theta_2 + b_{2j} \\
\pi_{3jk} &= \theta_3 + b_{3j} \\
\pi_4 &= \theta_4
\end{aligned}
$$

with distributional assumptions:

$$
\left(b_{0j}, b_{1j}, b_{2j}, b_{3j}\right) \sim \mathcal{N}\left(\mathbf{0},\ \mathbf{\Sigma}_b\right), \qquad c_{0k} \sim \mathcal{N}(0,\ \tau_{c_{00}})
$$

The covariate `prior_rt` is treated as having a fixed effect only (no random slope), consistent with Trott and Bergen's code.

In testing for cognitive warrant (H2), we are most concerned with the coefficient for `same`/`same_heading` which has mixed effects: $\pi_{2jk} = \theta_2 + b_{2j}$. Specifically, we are interested in the value and statistical significance of the fixed effect θ_2.

4.5.3. Modeling Strategy

Data from both experiments are analyzed using cross-classified mixed-effects models fitted in `lme4` (Bates et al., 2003), with `lmerTest` for Satterthwaite-approximated t-tests and p-values. Fixed-effect comparisons use likelihood ratio tests (LRT) on models fit by full maximum likelihood, not restricted maximum likelihood (REML). Under REML, the likelihood function is defined differently for models with different fixed effects, making their deviances non-comparable.

Average first-trial RT (`prior_rt`) is a control variable. Experiment 1 in the present study closely follows Trott and Bergen's primed sensibility judgment design; therefore, the primary mixed-effects analyses adopt their Experiment 2 strategy of controlling for sentence-specific baseline difficulty using the average first-trial response time (across all subjects) of the target sentence.

In alignment with Trott and Bergen, all analyses in Experiment 1 are of the target trial (i.e., the second sentence in each sentence pair), and analyses of RT include only correct responses.

Experiment 1

H1 and H2

```
model_h12_e1 <- lmer(log_rt ~ distance_bert + same_heading + prior_rt +
                       (1 + distance_bert + same_heading | subject) +
                       (1 | word),
                     control=lmerControl(optimizer="bobyqa"),
                     data = exp1_data, REML = FALSE)
```

H3: Natural spline interaction (primary test) The polynomial operationalization in my original analysis plan — a quadratic interaction between centered `distance_bert` and `same_heading` — conflates two distinct questions. The quadratic term `dist_c`^{2} tests whether *distance* affects `log_rt` in a nonlinear form, independently of heading congruence; the `dist_c`^{2}`:same_heading` interaction tests whether that nonlinear distance effect has a *different shape* between congruent and incongruent pairs. Neither term directly operationalizes H3's claim: that the *coefficient of* `same_heading` is itself a smooth, nonlinear function of `distance_bert`, peaking at intermediate distances.

The corrected specification models this directly by interacting `same_heading` with a natural spline basis expansion of `distance_bert`, fitted via `splines::ns()` (**?**). A natural spline with K interior knots represents the distance–RT relationship as a piecewise cubic polynomial constrained to be linear beyond the boundary knots, requiring $K+1$ basis columns. Interacting each basis column with `same_heading` allows the *categorical contrast between congruent and incongruent pairs* to vary smoothly and nonlinearly across the distance continuum without imposing a specific functional form such as a parabola. Three interior knots are placed at the 25th, 50th, and 75th percentiles of the observed `distance_bert` distribution, yielding four interaction terms and four degrees of freedom in the LRT. The LRT compares this model against the H2 baseline, in which `same_heading` enters only as a main effect (constant boundary coefficient across all distances):

```r
library(splines)
library(marginaleffects)

# Interior knot placement at quartiles
knots_h3 <- quantile(exp1_data$distance_bert,
                     probs = c(0.25, 0.50, 0.75), na.rm = TRUE)

# H2 baseline: same_heading effect constant across distance
model_h2_e1 <- lmer(
  log_rt ~ ns(distance_bert, knots = knots_h3) +
           same_heading +
           prior_rt +
```

```r
            (1 | subject) +
            (1 | word),
  data = exp1_data, REML = FALSE
)

# H3 model: same_heading effect varies smoothly with distance
model_h3_e1 <- lmer(
  log_rt ~ ns(distance_bert, knots = knots_h3) +
            same_heading +
            ns(distance_bert, knots = knots_h3):same_heading +
            prior_rt +
            (1 | subject) +
            (1 | word),
  data = exp1_data, REML = FALSE
)

# LRT: does the boundary effect vary with distance?
anova(model_h2_e1, model_h3_e1)

# Boundary effect profile: predicted same_heading contrast at each decile
deciles <- quantile(exp1_data$distance_bert,
                    probs = seq(0.1, 0.9, by = 0.1), na.rm = TRUE)

boundary_profile <- comparisons(
  model_h3_e1,
  variables = list(same_heading = 0:1),
  newdata   = datagrid(distance_bert = deciles,
                       prior_rt = mean(exp1_data$prior_rt, na.rm = TRUE)),
  re.form   = NA   # population-level predictions
)
```

H3 is supported if (1) the LRT is significant, rejecting the constant-boundary-effect H2 model in favor of the distance-varying H3 model; and (2) the `boundary_profile` shows the estimated congruent-minus-incongruent contrast at its largest absolute magnitude in the intermediate distance range (approximately 0.20–0.45 cosine units) and attenuating toward zero at both extremes. Both conditions are required: a significant LRT without an inverted-U profile shape would indicate that the boundary effect varies with distance in some other pattern, which the CI(d) framework does not predict.

Random effects structure. An initial model including by-subject random slopes for `same_heading` produced a singular fit: the random slope variance was estimated at

7.5×10^{-6} with a boundary correlation of -1.00, indicating the slope is not identifiable at the pilot sample size of 25 words and 187 subjects. Both models are therefore fit with random intercepts only, following the principle of using the maximal structure supported by the data (Barr et al., 2013). The by-subject random slope for `same_heading` will be reinstated in the full confirmatory study.

H3 robustness check: generalized additive model (GAM) A GAM fitted via `mgcv` (**?**) serves as the nonparametric robustness check. The GAM fits a separate smooth function for `distance_bert` within each level of `same_heading` (`by = same_heading_f`), with random intercepts for subjects and words approximated as random-effect smooths. Convergence of the GAM and natural spline interaction on the same direction and peak location constitutes evidence that the H3 finding is not an artifact of the spline parameterization.

```r
library(mgcv)
library(gratia)

exp1_data <- exp1_data %>%
  mutate(same_heading_f = factor(same_heading,
                                 levels = c(0, 1),
                                 labels = c("Incongruent", "Congruent")))

model_h3_gam <- gam(
  log_rt ~ s(distance_bert, by = same_heading_f, k = 5) +
           same_heading_f +
           prior_rt +
           s(subject, bs = "re") +
           s(word, bs = "re"),
  data   = exp1_data,
  method = "REML"
)

# Smooth difference: does the Congruent minus Incongruent contrast
# peak at intermediate distances?
difference_smooths(model_h3_gam,
                   smooth   = "s(distance_bert)",
                   by_var   = "same_heading_f")
```

H4: hierarchical distance (within same_nbhd = 1 only)

```
model_h4_e1 <- lmer(log_rt ~ distance_bert + hier_distance + prior_rt +
(1 + distance_bert + hier_distance | subject) +
(1 | word),
data = exp1_data %>% filter(same_nbhd == 1),
REML = FALSE)
```

H5: `ambiguity_type` × `same_heading`

```
model_h5_e1 = lmer(log_rt ~ distance_bert + same_heading * ambiguity_type +
                                        prior_rt +
                    (1 + distance_bert + same_nbhd + ambiguity_type | subject)
                    ↪  +
                    (1 | word),
                  control=lmerControl(optimizer="bobyqa"),
                  data = exp1_data,
                  REML = FALSE)
```

For H6, `hier_distance` is added to test for mediation.

Experiment 2

First, general notes applying across all models:

1. On variable centering. `distance_bert_foil`, `distance_bert_target`, and `hier_distance` should all be mean-centered before entry, both to aid convergence and to make the intercept in E2-H1 interpretable as the expected `delta_score` at average distance values.
2. On the LRT sequence. The hypothesis-to-model comparison structure mirrors Experiment 1: each model is compared to the one immediately below it in complexity via `anova()` with `REML = FALSE`, consistent with the likelihood ratio test approach.
3. The covariate `prior_rt` was necessary in Experiment 1 to compensate for the inherent difference in sentence difficulties which would presumably affect a subject's response time. Trials in Experiment 2 are not timed. It *is* possible that the degree of sentence difficulty may affect appropriateness ratings, requiring perhaps some function of the two stimulus sentences' `prior_rt` values. It is my judgment

that the average-sentence-difficulty and appropriateness-rating relationship is not significant. If anything, simpler sentences might get higher appropriateness ratings; however, Experiment 2 will not include the covariate `prior_rt` or a proxy for sentence-pair "inherent appropriateness" at this time.

E2-H1: Target superiority. This is a mean-level test (is `delta_score` > 0 on average?). It requires an intercept-only model, with the intercept tested against zero:

```r
model_h1_e2 <- lmer(delta_score ~ 1 + (1 | subject) + (1 | word),
                    control = lmerControl(optimizer = "bobyqa"),
                    data = exp2_data,
                    REML = FALSE)

# Test whether intercept differs from zero
summary(model_h1_e2)  # examine fixed intercept and CI
confint(model_h1_e2, parm = "(Intercept)")
```

E2-H2: Cross-category foil disadvantage. E2-H2 is the analog of Experiment 1's H2 and tests whether the categorical position of the foil (i.e., whether `same_nbhd` $= 0$ or 1) significantly affects the appropriateness rating independent of semantic distance.

```r
model_h2_e2 <- lmer(delta_score ~ distance_bert + same_nbhd +
                    (1 + distance_bert + same_nbhd | subject) +
                    (1 | word),
                    control = lmerControl(optimizer = "bobyqa"),
                    data = exp2_data %>% filter(rating_type == "foil\_same" |
                    ↪  "foil\_cross"),
                    REML = FALSE)
```

E2-H3: Continuous semantic fit. E2-H3 requires controlling for target-heading distance which means that the model needs two `distance_bert` predictors: *foil-to-sentence-pair distance* and *target-to-sentence-pair distance*. Thus, the model has the following predictors:

- `distance_bert_target`: cosine distance between the sentence-pair representation and the target heading
- `distance_bert_foil`: cosine distance between the sentence-pair representation and the foil heading

```
model_h3_e2 = lmer(delta_score ~ distance_bert_foil + distance_bert_target +
                   same_heading +
                   (1 + distance_bert_foil + distance_bert_target +
                   same_heading | subject) +
                   (1 | word),
                   control = lmerControl(optimizer = "bobyqa"),
                   data = exp2_data,
                   REML = FALSE)
```

Some things to note with two `distance_bert` predictors:

On the random effects structure. Adding a second distance predictor to the by-subject random slopes makes the model more complex and may not converge. Following Barr et al. (2013) as Trott and Bergen did, I will start "maximal" and reduce if needed — the first candidate for removal would be `distance_bert_target` from the random slopes, since target-heading distance is constant within item across subjects and may contribute little individual variation.

On the theoretical roles of the two predictors. `distance_bert_foil` is the primary predictor of interest for E2-H3 – its coefficient tests whether more distant foils produce larger discrimination. `distance_bert_target` is a control covariate, not a hypothesis in itself. In other words, `distance_bert_foil` is confirmatory and `distance_bert_target` is a nuisance control.

On variable construction. Both distances are computed against the same sentence-pair representation (mean pooling of the two sentence embeddings), so they share a common reference vector. This means `distance_bert_foil` and `distance_bert_target` will likely be moderately correlated — items where the target heading is distant from the sentence pair tend also to have distant foils. The variance inflation factor will need to be checked before interpreting individual coefficients.

E2-H4: Structural moderation (within same_nbhd = 1 only). E2-H4 adds `hier_distance` as a predictor controlling for both distance terms:

```
model_h4_e2 <- lmer(delta_score ~ hier_distance +
                    distance_bert_foil + distance_bert_target +
                    (1 + hier_distance + distance_bert_foil +
                    distance_bert_target | subject) +
                    (1 | word),
                    control = lmerControl(optimizer = "bobyqa"),
                    data = exp2_data %>% filter(same_nbhd == 1),
                    REML = FALSE)
```

Note that `hier_distance` here is the path distance between the foil heading and the target heading in the LCSH DAG — not between either heading and a sentence representation.

H7: Cross-Experiment Convergence

```
# Compute item-level implicit disruption index from Exp 1

disruption_by_item <- exp1_data %>%
group_by(word) %>%
summarise(
rt_diff_sense = mean(log_rt[same_heading == 0], na.rm = TRUE),
rt_same_sense = mean(log_rt[same_heading == 1], na.rm = TRUE),
disruption_index = rt_diff_sense - rt_same_sense
)

# Merge with item-level mean delta_r from Exp 2

h7_data <- disruption_by_item %>%
left_join(exp2_data %>% group_by(word) %>%
summarise(mean_delta_r = mean(delta_r, na.rm = TRUE)))
```

```r
# H7: positive item-level correlation

cor.test(h7_data$disruption_index, h7_data$mean_delta_r,
method = “pearson”, alternative = “greater”)
```

4.6. Chapter Summary

This chapter has described a five-phase methodology operationalizing cognitive warrant through (1) systematic LCSH mapping of ambiguous stimulus words; (2) LCSH DAG construction and structural variable computation; (3) two behavioral experiments replicating and extending Trott and Bergen's (2023) study; (4) BERT embedding of context sentences and subject headings; and (5) cross-classified mixed-effects modeling. These methods facilitate answering the supporting research questions in Section 1.3.2 and determining to what extent knowledge organization systems such as LCSH can be evaluated as representations of semantic structure using computational semantic similarity measures and behavioral validation.

CHAPTER 5: PRELIMINARY RESULTS

5.1. Status of study

As of mid-June 2026, application #2446127-1 for exempt certification was still in the review process by the University of Denver Institutional Review Board. This precluded, prior to finishing the capstone course, (1) expert consultant validation of LCSH stimulus mapping by two librarians with cataloging experience, and (2) pilot testing of Experiments 1 and 2 with SONA participants via the Pavlovia platform. In other words, this study currently has no original results.

I have, however, conducted a proof of concept to demonstrate the plausibility of cognitive warrant. A secondary analysis of Trott and Bergen's Experiment 2 behavioral data demonstrates that the analytic pipeline is end-to-end executable on real linguistic material. The results of this analysis and their implications for the full study are discussed.

5.2. The Necessity of Behavioral Data for Cognitive Information Values

Cognitive information and its derived metrics were defined in the Theoretical Framework chapter. In Section 3.5.5, the source of three variables used in those metrics—Y, d, S—was distinguished from that of the probabilities $P(Y_i \mid d_i)$ and $P(Y_i \mid d_i, S_i)$. The values for Y_i, d_i, S_i are pre-experimental in that sense identity, semantic distance, and heading congruence are determinable before any study participant responds. The probabilities, however, are estimated from behavioral data.

They are item-level fitted values from mixed-effects models trained on participant response times. The theoretical quantities of cognitive information $\mathrm{CI}(d)$, warrant efficiency η, expected cognitive information $\overline{\mathrm{CI}}$, and pointwise cognitive information pCI_i are thus produced from fitted regression models. Yet, they all trace back to two regression equations and one entropy function.

5.3. The Analytic Pipeline: Logistic Regression and the Estimation of Cognitive Warrant Metrics

The preliminary results reported in this chapter are drawn from a secondary analysis of the stimulus materials from Trott and Bergen(2023), Experiment 2, merged with a newly constructed mapping of those materials onto Library of Congress Subject Headings. This merge is recorded in `s2_trials_lcsh_distances.csv` and constitutes the analytic dataset for this chapter. It is important to state at the outset what this proof of concept does, and does not, establish. It does not test whether human semantic cognition is sensitive to LCSH categorical boundaries—that is the purpose of Experiment 1, proposed in Chapter 3.

What it does is retroactively fit the LCSH mapping onto an existing, published stimulus set and carry that mapping through the full computational pipeline used to derive the cognitive warrant metrics defined in Chapter 4: the fitted logistic models, the conditional entropies, the piecewise and pointwise cognitive information quantities, and the resulting system-level warrant score. Because Trott and Bergen's stimuli already carry an independently established ground-truth sense classification (same-sense vs. different-sense, per dictionary annotation) and an existing BERT cosine distance measure, this dataset offers exactly the two ingredients—a binary outcome and a continuous distance predictor—needed to exercise the pipeline on real linguistic material, without requiring new data collection from human participants.

Any patterns reported below therefore speak to the soundness of the methodology, the analytic pipeline, and the behavior of the independent variables under realistic conditions; they are preliminary and illustrative, not confirmatory, evidence for cognitive warrant itself.

5.3.1. From Trial-Level Records to the Analytic Sample

`s2_trials_lcsh_distances.csv` is constructed at the trial level, inheriting the repeated-measures structure of Trott and Bergen's Experiment 2 (multiple subjects responding to each sentence pair). The quantities of interest in this proof of concept pipeline, however, are properties of the *stimulus pair* rather than of any individual trial: the ground-truth sense-identity label Y_i, the BERT cosine distance d_i, the LCSH heading congruence indicator S_i (`same_heading`), and the hierarchical distance h_i (`hier_distance`). The first step of the pipeline is therefore a collapse from trials to unique stimulus pairs: `s2_trials_lcsh_distances.csv` is grouped by item (sentence-pair identifier), retaining one row per pair with its associated Y_i, d_i, S_i, and h_i. This collapsed, pair-level table is the input to the logistic regressions in Section 5.3.3.

```r
# Step 0: collapse trial-level records to unique stimulus pairs
pairs_df <- s2_trials_lcsh_distances %>%
  distinct(pair_id, same, distance_bert, same_heading, hier_distance)
```

5.3.2. Why Logistic Regression

The outcome $Y_i \in \{0, 1\}$ is binary, and every downstream quantity in the pipeline—conditional entropy, cognitive information, pointwise cognitive information—is defined in terms of a *probability* $P(Y_i \mid \cdot)$, not a raw score. This rules out ordinary least squares: an OLS fit of Y_i on d_i would happily return predicted values below 0 or above 1, which cannot be interpreted as probabilities and cannot be substituted into an entropy function that requires $p \in [0,1]$. Logistic regression is the natural alternative because it models the log-odds of Y_i as a linear function of the predictors,

$$\operatorname{logit} P(Y_i = 1 \mid d_i) \equiv \ln \frac{P(Y_i = 1 \mid d_i)}{P(Y_i = 0 \mid d_i)} = \beta_0 + \beta_1 d_i,$$

and recovers a probability, guaranteed to lie in the interval $[0,1]$, by inverting the logit with the logistic (sigmoid) function $\sigma(\cdot) = \left(1 + e^{-(\cdot)}\right)^{-1}$.

This choice is not merely a technical convenience for keeping fitted values in range. It is the direct empirical counterpart of the distributional assumption used in Chapter 4 to derive H3 as a mathematical theorem rather than an inductive conjecture: that assumption was $P(Y = 1 \mid d) = \sigma(\alpha - \beta d)$ for $\alpha, \beta > 0$—exactly the functional form that a logistic regression of Y_i on d_i estimates. Fitting the baseline model is thus not an arbitrary modeling decision layered on top of the theory; it is the theory's own assumed function, estimated from data. The same logic extends to the full model: because $\mathrm{CI}(d) \equiv I(Y; S \mid d)$ was identified as conditional mutual information, and the null hypothesis $\mathrm{CI}(d) = 0$ is equivalent to $Y \perp S \mid d$, the likelihood-ratio test comparing the full logistic model against the baseline is a finite-sample estimator of the sign and significance of $\mathrm{CI}(d)$: rejecting $\beta_S = 0$ in the regression is formally equivalent to rejecting $I(Y; S \mid d) = 0$ in the information-theoretic formulation. Logistic regression is therefore the load-bearing link between the regression framework and the information-theoretic one, not an incidental estimation choice between two otherwise unrelated frameworks.

Finally, fitted logistic probabilities are preferable to raw empirical proportions (e.g., binning d and computing the observed fraction of same-sense pairs per bin) as the input to the entropy calculations. Raw proportions are noisy wherever a bin contains few pairs, can equal exactly 0 or 1 (making the entropy function degenerate), and depend on an arbitrary choice of bin width. A fitted logistic curve borrows strength across the full range of d_i, producing smooth, stable probability estimates even in sparsely populated regions of the distance continuum—which matters directly for this proof of concept, since Trott and Bergen's materials were not sampled to guarantee dense coverage of the intermediate distance range that the cognitive warrant framework identifies as theoretically critical.

5.3.3. Step 1: Fitting the Baseline and Full Models

Two nested logistic regressions are fit to the collapsed pair-level data. The **baseline model** predicts sense identity from continuous distance alone:

$$\operatorname{logit} P(Y_i = 1 \mid d_i) = \beta_0 + \beta_1 d_i. \tag{5.1}$$

The **full model** adds LCSH heading congruence:

$$\operatorname{logit} P(Y_i = 1 \mid d_i, S_i) = \gamma_0 + \gamma_1 d_i + \gamma_2 S_i. \tag{5.2}$$

(A structural extension replacing S_i with the joint signature (S_i, h_i), corresponding to $\mathrm{CI}_{\mathrm{struct}}(d)$ in Chapter 3, is deferred to the fully powered study; the two-model comparison above is sufficient to exercise the core pipeline.)

```
# Step 1: baseline and full logistic models on the pair-level data
m_baseline <- glm(same ~ distance_bert,
                  data = pairs_df, family = binomial(link = "logit"))
```

```r
m_full <- glm(same ~ distance_bert + same_heading,
              data = pairs_df, family = binomial(link = "logit"))

# Likelihood-ratio test: the regression-side estimate of CI(d) > 0,
# i.e., the finite-sample test of I(Y; S | d) = 0
anova(m_baseline, m_full, test = "Chisq")

# Fitted probabilities: P(Y_i = 1 | d_i) and P(Y_i = 1 | d_i, S_i)
pairs_df <- pairs_df %>%
  mutate(
    p_baseline = predict(m_baseline, type = "response"),
    p_full     = predict(m_full,     type = "response")
  )
```

5.3.4. Step 2: From Fitted Probabilities to Conditional Entropy

Each fitted probability is converted to an entropy via the binary entropy function

$$h_b(p) = -p \log_2 p - (1-p)\log_2(1-p),$$

applied item-by-item to the two sets of fitted probabilities:

$$\widehat{H}(Y_i \mid d_i) = h_b\big(\hat{P}(Y_i = 1 \mid d_i)\big), \tag{5.3}$$

$$\widehat{H}(Y_i \mid d_i, S_i) = h_b\big(\hat{P}(Y_i = 1 \mid d_i, S_i)\big). \tag{5.4}$$

Equation 5.3 is the model-implied disambiguation load carried by distance alone for pair i; Equation 5.4 is the residual disambiguation load once the LCSH heading assignment is also known. The difference between the two, evaluated pair by pair, is the raw material for both the piecewise and pointwise cognitive information quantities below.

```
# Step 2: binary entropy of the fitted probabilities
h_b <- function(p) {
  p <- pmin(pmax(p, 1e-12), 1 - 1e-12)  # guard against log(0)
  -p * log2(p) - (1 - p) * log2(1 - p)
}

pairs_df <- pairs_df %>%
  mutate(
    H_baseline = h_b(p_baseline),
    H_full     = h_b(p_full)
  )
```

5.3.5. Step 3: Piecewise Cognitive Information

CI(d) is a function of the continuous variable d, but any single dataset only samples d at a finite, unevenly distributed set of points. Two complementary estimates of CI(d) are computed from the fitted models. The first is a *smooth* curve, obtained by evaluating the full model's fitted probability at both $S = 0$ and $S = 1$ across a fine grid of d values spanning the observed range, then combining the two curves via the locally estimated mixture weight $\hat{\pi}(d) = \hat{P}(S = 1 \mid d)$ (a fixed-window empirical average around each grid point); this is the curve implied directly by the logistic functional form and is the one plotted against the theoretical prediction of Chapter 3 (an inverted-U, concave in d, vanishing at both extremes).

The second is the **piecewise cognitive information** function: the observed range of d is partitioned into K contiguous, non-overlapping bins. For bin k with midpoint x_k, containing pairs $\{i : d_i \in \text{bin } k\}$, let $\bar{p}_{\text{full},k} = \frac{1}{n_k} \sum_{i \in k} \hat{P}(Y_i = 1 \mid d_i, S_i)$

be the bin's average fitted probability under the full model. Then

$$\widehat{\mathrm{CI}}_k = h_b\big(\bar{p}_{\mathrm{full},k}\big) \; - \; \frac{1}{n_k}\sum_{i\in k} \widehat{H}(Y_i \mid d_i, S_i), \tag{5.5}$$

where n_k is the number of pairs in bin k and h_b is the binary entropy function of Equation 5.3. Equation 5.5 is a direct application of Jensen's inequality to the concavity of h_b: the first term is the entropy *of the mixture* $\bar{p}_{\mathrm{full},k}$, and the second is the *mixture of entropies* of that same set of pair-level probabilities, so $\widehat{\mathrm{CI}}_k \geq 0$ is guaranteed algebraically, not merely expected empirically.[6] Equation 5.5 also plays a specific methodological role that the smooth curve cannot: it does not depend on the logistic curve's behavior being correctly extrapolated far from the data, and it can be recomputed using raw empirical proportions in place of fitted probabilities as a fully nonparametric cross-check. Agreement between the smooth, model-implied $\mathrm{CI}(d)$ curve and the piecewise, bin-wise $\widehat{\mathrm{CI}}_k$ estimate is itself a diagnostic: it indicates that the inverted-U prediction is not an artifact of the assumed logistic functional form, but is recoverable directly from the empirical distribution of outcomes within each region of the distance continuum.

```
# Step 3: piecewise (binned) cognitive information
K <- 6  # number of equal-width distance bins; sensitivity-checked
pairs_df <- pairs_df %>%
  mutate(distance_bin = cut(distance_bert, breaks = K))


ci_piecewise <- pairs_df %>%
```

[6] An earlier implementation substituted h_b of a fitted probability from a model estimated independently of the full model (i.e., using only d, ignoring S) in place of $h_b(\bar{p}_{\mathrm{full},k})$. Because that independent model is not guaranteed to reproduce the full model's own S-marginalized mixture probability, the resulting quantity is not guaranteed non-negative by Jensen's inequality — a violation that surfaced as a small, easily-dismissed sign flip in one sparse bin under the S model, but as a sustained, substantial negative region under the h^{finite} model (Section 5.4.3), which is what prompted the correction reported here.

```r
group_by(distance_bin) %>%
summarise(
  n_pairs  = n(),
  x_mid    = mean(distance_bert),
  # entropy of the bin's average P(Y=1|d,S) -- NOT the entropy of an
  # independently-fit baseline model's prediction; see Eq. above / footnote
  H_bar_d  = h_b(mean(p_full)),
  H_bar_dS = mean(H_full),
  CI_k     = H_bar_d - H_bar_dS,   # >= 0 by Jensen's inequality
  P_d      = n_pairs / nrow(pairs_df)
)
```

5.3.6. Step 4: Pointwise Cognitive Information

Where the piecewise function characterizes disambiguation gain at the level of a distance region, *pointwise cognitive information* characterizes it at the level of a single stimulus pair, using the fitted probability of whichever outcome was actually observed for that pair:

$$\mathrm{pCI}_i = \log \frac{\hat{P}(Y_i \mid d_i,\, S_i)}{\hat{P}(Y_i \mid d_i)}. \tag{5.6}$$

A positive pCI_i indicates that the LCSH heading assignment for pair i moved the model's posterior toward the correct sense-identity outcome, relative to distance alone; a value near zero indicates the heading assignment was informationally redundant with distance for that pair; a negative value indicates the heading assignment actively misled the model away from the correct outcome. Because $\mathbb{E}[\mathrm{pCI}_i \mid d_i \in \text{bin } k] = \widehat{\mathrm{CI}}_k$ by construction, computing the mean pointwise value within each bin and comparing it to the piecewise estimate from Equation 5.5 serves as an internal consistency check on the pipeline: the two quantities are computed from different formulas and different levels of aggregation, and their agreement (up to sampling variability) is evidence that the implementation, not merely the underlying theory, is behaving as specified.

This identity holds only when $\hat{P}(Y_i \mid d_i)$ in Equation 5.6 and the marginal term in $\widehat{\mathrm{CI}}_k$ (Equation 5.5) are the *same* quantity. Both are therefore estimated the same way: as $\bar{p}_{\text{full},k}$, the bin-mean fitted probability under the full model, rather than from an independently-fit model of d alone.[7]

```
# Step 4: pointwise cognitive information, evaluated at the observed Y_i.
# p_bar_full comes from ci_piecewise (Step 3) -- the same bin-mean full-
# model probability CI_k is built from, not an independently-fit model.
pairs_df <- pairs_df %>%
  left_join(ci_piecewise %>% select(distance_bin, p_bar_full),
            by = "distance_bin") %>%
  mutate(
    p_full_obs     = ifelse(same == 1, p_full, 1 - p_full),
    p_marginal_obs = ifelse(same == 1, p_bar_full, 1 - p_bar_full),
    pCI            = log2(p_full_obs / p_marginal_obs)
  )

# Internal consistency check: mean(pCI) within a bin should
# approximate that bin's piecewise CI_k
pairs_df %>%
  group_by(distance_bin) %>%
  summarise(mean_pCI = mean(pCI)) %>%
  left_join(ci_piecewise, by = "distance_bin")
```

Ranking pairs by pCI_i additionally previews the heading-reform diagnostic developed in Chapter 3: the pairs with the most negative pointwise values in this

[7] An earlier implementation used an independently-fit baseline model (predicting Y from d alone) for this term. Because that baseline model is not guaranteed to reproduce the full model's own bin-mean probability, the two quantities compared by the consistency check were not built from the same reference, and their disagreement in sparse bins reflected the baseline model's local misspecification (formally, a KL-divergence term) rather than sampling noise. Using $\bar{p}_{\text{full},k}$ for both removes that source of bias, leaving only genuine finite-sample variance – see Section 5.4.1 for the resulting reduction in the consistency check's largest bin-level gaps.

proof-of-concept sample are LCSH-mapped items whose heading assignment, applied to Trott and Bergen's existing stimulus set, would misdirect a purely distributional judgment about sense identity. Surfacing such items here is a demonstration that the diagnostic is computable end-to-end on real material; it is not, on the strength of a secondary analysis of an existing stimulus set, a claim that any specific LCSH heading should be revised.

5.3.7. Step 5: Expected Cognitive Information and Warrant Efficiency

The piecewise table is aggregated into the two scalar, system-level metrics defined in Chapter 3. The *expected cognitive information* is the bin-weighted average of the piecewise estimates,

$$\overline{\mathrm{CI}} = \sum_{k=1}^{K} P(d \in \mathrm{bin}\ k) \cdot \widehat{\mathrm{CI}}_k, \tag{5.7}$$

and *warrant efficiency* normalizes this by the corresponding weighted average of the baseline disambiguation load,

$$\bar{\eta} = \frac{\overline{\mathrm{CI}}}{\sum_{k=1}^{K} P(d \in \mathrm{bin}\ k) \cdot \overline{H}_k(Y \mid d)}. \tag{5.8}$$

$\overline{\mathrm{CI}}$ and $\bar{\eta}$ are the two headline numbers this proof of concept produces: a bits-valued estimate of how much uncertainty about sense identity the LCSH mapping resolves on average across Trott and Bergen's stimulus distribution, and the proportion of the available disambiguation load that this resolution represents.

```r
# Step 5: expected cognitive information and warrant efficiency
CI_bar  <- with(ci_piecewise, sum(P_d * CI_k))
eta_bar <- CI_bar / with(ci_piecewise, sum(P_d * H_bar_d))
```

5.3.8. Scope of the Proof of Concept

Three things follow from a successful run of the pipeline described above, and one thing does not. First, it demonstrates that the LCSH mapping procedure—linking a lexicographically defined sense pair to an LCSH authority record, and computing `same_heading` and `hier_distance` from that mapping—can be executed against an existing, independently collected stimulus set, rather than only against materials purpose-built for this study. Second, it demonstrates that the logistic-regression-to-entropy machinery produces well-behaved, interpretable output (non-degenerate fitted probabilities, non-negative $\widehat{\mathrm{CI}}_k$ estimates, an internally consistent correspondence between pCI_i and $\widehat{\mathrm{CI}}_k$) when applied to real linguistic material at the scale of a full stimulus set, not only to a hand-worked toy case. Third, it provides an initial, if uncontrolled, read on the statistical behavior of the key independent variables—most importantly, whether `same_heading` exhibits enough independent variation from `distance_bert` to avoid the redundancy bound $\mathrm{CI}(d) \leq H(S \mid d)$ collapsing toward zero, which would indicate that the fully powered study is attempting to detect an effect that cannot exist for structural reasons, independent of statistical power.

What does not follow is any claim about cognitive warrant itself. The pair-level outcome Y_i in this analysis is a lexicographic, dictionary-based classification of sense identity, not a measurement of human behavior; the pipeline exercised here never asks whether a person's processing of these sentence pairs is affected by the LCSH boundary, only whether the boundary and the BERT distance jointly track a symbolic, pre-existing sense classification. That is a necessary but not sufficient condition for cognitive warrant: a boundary that is uninformative about lexicographic sense identity is very unlikely to be cognitively meaningful either, but a boundary that is informative about lexicographic sense identity has not thereby been shown to correspond to any discontinuity in human semantic processing. That inferential step is reserved for Experiment 1.

5.4. Results and Implications for Cognitive Warrant

5.4.1. Model Comparisons and the Shape of CI(d)

The baseline model, $\operatorname{logit} P(Y_i = 1 \mid d_i) = \beta_0 + \beta_1 d_i$, fit cleanly and confirms the expected relationship between continuous semantic distance and ground-truth sense identity in Trott and Bergen's materials: $\hat{\beta}_0 = 1.563$ ($SE = 0.310$, $p < .001$), $\hat{\beta}_1 = -9.744$ ($SE = 1.351$, $p < .001$), reducing residual deviance from 381.10 to 299.94 on 297 df. Two full models were fit against this baseline — one adding LCSH heading congruence (S_i, `same_heading`), one adding hierarchical connectivity (h_i^{finite}, whether prime and target share any BT/NT ancestor within the search horizon) — and they behave very differently, for a reason that is itself the central finding of this chapter.

Table 5.1 summarizes all three models. The S-augmented model could not be estimated by ordinary maximum likelihood: `glm()` returned coefficients an order of magnitude beyond interpretability ($\hat{\beta}_S = 53.1$, $SE = 49{,}320$) and residual deviance collapsed from 299.94 to 1.7×10^{-9} on the one added degree of freedom — a signature of complete separation, not a strong effect. Firth's penalized-likelihood refit (`logistf`) recovers finite, interpretable coefficients ($\hat{\beta}_S = 10.363$, profile 95% CI $[7.70, 22.47]$), but the profile-likelihood χ^2 for S is reported as infinite ($p = 0$): the likelihood surface has no interior maximum with respect to β_S, only a penalized approximation to one. The h^{finite}-augmented model shows the same pathology in a milder form: `glm()` gives $\hat{\beta}_{h^{\text{finite}}} = 19.29$ with $SE = 937.0$ ($z = 0.02$, $p = .98$), the same large-estimate/large-SE/near-zero-z signature, while the likelihood-ratio test comparing this model to baseline is well-behaved ($\Delta\text{Dev} = 89.43$ on 1 df, $p < .001$) because the LRT compares log-likelihoods directly and does not depend on the asymptotic normality that the Wald test requires. Firth's refit again gives a finite, moderate estimate ($\hat{\beta}_{h^{\text{finite}}} = 5.303$, 95% CI $[3.34, 10.15]$).

Table 5.1: Logistic regression coefficients, three models predicting ground-truth sense identity Y_i from BERT cosine distance d_i and LCSH structural variables. Firth (penalized ML) rows are the estimates used in the entropy calculations that follow. Standard errors in parentheses where finite and interpretable.

Model	Intercept	d	S / h^{finite}	Note
Baseline (glm)	1.563 (.310)	−9.744 (1.351)	—	Converged normally
Full, S (glm)	−26.57 (54,150)	≈ 0 (150,200)	53.13 (49,320)	Complete separation
Full, S (Firth)	−4.472	−4.956	10.363	$\chi^2_S = \infty$, $p = 0$
Full, h^{finite} (glm)	−17.61 (937.0)	−6.407 (1.211)	19.29 (937.0)	Quasi-separation
Full, h^{finite} (Firth)	−3.657	−6.265	5.303	LRT $\chi^2(2)$=164.75, $p < .001$

The piecewise cognitive information curve for the S-based model (Table 5.2) is not a simple inverted-U over its six bins: it rises from $\widehat{\text{CI}} \approx 0.72$ bits in the lowest bin to a maximum of $\widehat{\text{CI}} \approx 0.89$ bits at $d \approx 0.19$ (bin 2), falls to essentially zero ($\widehat{\text{CI}} \approx 0.00002$ bits) by $d \approx 0.40$ (bin 4), then rises again to $\widehat{\text{CI}} \approx 0.56$–$0.71$ bits in the two sparsest, highest-distance bins ($n = 26$, $n = 16$). The smooth, model-implied curve peaks higher and at a slightly lower distance than any single piecewise bin (≈ 0.950 bits near $d \approx 0.168$), consistent with the coarse six-bin discretization smoothing over a sharper underlying peak; whether the smooth curve also exhibits the piecewise table's late-bin upturn, or declines more smoothly through that sparse region, has not been checked here and would require inspecting `grid_d$CI_grid_v2` directly rather than just its maximum. The late-bin upturn is plausibly connected to S's complete separation from Y (Section 5.4.2): the Firth-penalized fit's entropy estimates are most sensitive to exactly this kind of small-n instability where the underlying model is degenerate.

It does not, however, approach zero at the low-distance extreme within this stimulus set's range: even in the lowest bin ($\bar{d} = 0.089$), $\overline{H}(Y \mid d) = 0.909$ bits — the baseline model assigns only 67–78% confidence to same-sense-hood at the smallest distances Trott and Bergen's materials contain, not the near-certainty the theoretical curve assumes at $d \to 0$. This is a property of the stimulus set (originally constructed to test dictionary sense boundaries, not sampled to cover the full theoretical range of d) rather than a pipeline defect, and it means this proof of concept observes (an admittedly non-monotonic version of) the descending half of the predicted curve, but cannot speak to the ascending, near-zero-distance half.

Table 5.2: Piecewise cognitive information, S-based full model. $n = 299$ pairs.

Bin	n_k	$\widehat{\mathrm{CI}}_k$ (bits)	$P(d \in k)$	Contribution
$[0.028, 0.136]$	72	0.723	0.241	0.174
$(0.136, 0.243]$	74	0.890	0.247	0.220
$(0.243, 0.350]$	79	0.517	0.264	0.137
$(0.350, 0.458]$	32	0.0000	0.107	0.000
$(0.458, 0.565]$	26	0.562	0.087	0.049
$(0.565, 0.673]$	16	0.712	0.054	0.038
			$\overline{\mathrm{CI}}_S$	**0.618 bits**
			$\bar{\eta}_S$	**0.938**

The internal consistency check (mean pointwise pCI_i against the piecewise $\widehat{\mathrm{CI}}_k$) now shows close agreement across all six bins: differences of 0.03–0.05 bits in the four best-populated bins (1–3, 5) and 0.03 and 0.08 bits in the two sparsest, highest-distance bins (4, 6) — a substantial reduction from the 0.14–0.85-bit gaps reported under the earlier pipeline, where pCI_i and $\widehat{\mathrm{CI}}_k$ were built from different marginal-probability references (Section 5.3.6). Aggregated across all bins, the unweighted mean pCI (0.651) is now only about 5% above $\overline{\mathrm{CI}}_S$ (0.618 bits), down from the ~16% gap reported previously. The residual bin-level variation that remains is consistent with ordinary finite-sample noise in a per-pair log-probability-ratio statistic rather than a systematic estimation artifact.

The hierarchical-distance model's equivalent aggregate check is tighter still: unweighted mean pCI_h (0.193) against $\overline{\mathrm{CI}}_h$ (0.190) is a ~1.6% gap, effectively exact agreement.

5.4.2. The Perfect Correspondence Between S and Y: Artifact or Finding?

The single most consequential empirical fact in this proof of concept is that $S_i = Y_i$ for all 299 stimulus pairs, without exception, across both homonymous ($n = 83$) and polysemous ($n = 216$) items. This is why the S-augmented model could not converge by ordinary maximum likelihood: knowing S_i is logically equivalent to knowing Y_i, so $H(Y \mid d, S) \to 0$ and $\mathrm{CI}(d) \to H(Y \mid d)$ at its ceiling everywhere, giving the corrected warrant efficiency $\bar{\eta}_S = 0.938$ (Table 5.2).[8] That number does not mean LCSH heading congruence resolves 94% of the disambiguation load in any empirically interesting sense; it means the variable, as constructed here, is a re-encoding of the outcome it is being used to predict (see Figure 5.1).

Is this avoidable? Yes, and identifying why it occurred here points directly at the fix. Each of Trott and Bergen's two dictionary senses per word (M1,

[8]This figure changed from the previously reported 0.944 under the correction described in Section 5.3.5; the qualitative point – S's near-ceiling efficiency is a mechanical consequence of construction, not an empirical finding – is unaffected.

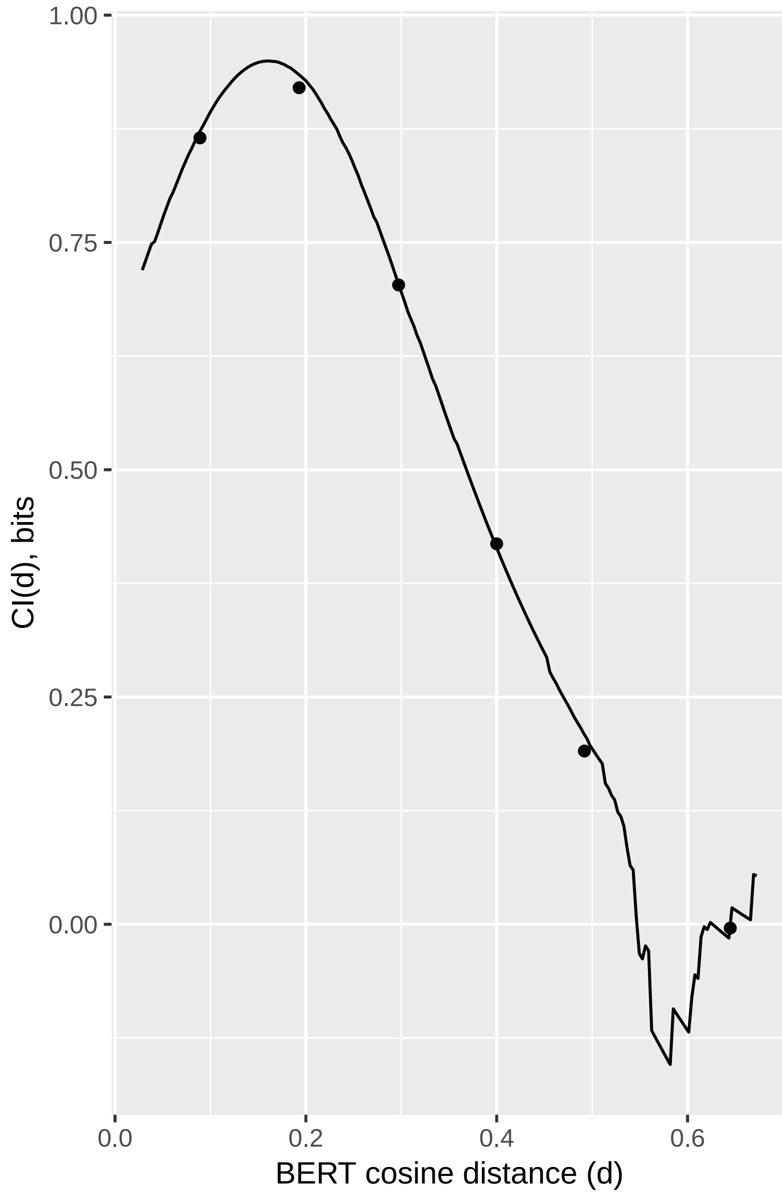


Figure 5.1: Cognitive information CI(d) as a function of BERT cosine distance d (Firth-fit logistic; solid curve), computed as the entropy of the full model's own S-marginalized mixture probability minus the mixture of its conditional entropies (Eq. 5.5), guaranteeing non-negativity by Jensen's inequality. Points mark the six binned estimates $\widehat{\mathrm{CI}}_k$ from Table 5.2 at each bin's mean distance. CI(d) peaks at ≈ 0.950 bits near $d \approx 0.168$; the piecewise estimate (Table 5.2) is not strictly monotonic beyond this peak, dipping toward zero near $d \approx 0.40$ before rising again in the two sparsest, highest-distance bins – see Section 5.4.1 for discussion. The curve does not reach zero at the low-distance extreme because the baseline model retains residual uncertainty at the smallest distances in Trott and Bergen's materials. As noted in Section 5.4.2, $S \equiv Y$ in this secondary dataset, making the magnitude shown a mechanical ceiling rather than an empirical estimate of LCSH's cognitive information content; see Section 5.4.3 for the non-tautological hierarchical-distance result.

M2) was mapped onto exactly one LCSH authority record, invariant across the two example sentences instantiating that sense (`heading_uri_prime` depends only on `sense_uid_prime`, not on the specific sentence). That is a *sense-anchored* mapping procedure: the analyst, already knowing which dictionary sense a given sentence instantiates, selected the single best-fitting LCSH heading for that sense as a whole. Under this procedure, $S_i = 1$ if and only if prime and target were drawn from the same dictionary sense — which is the definition of $Y_i = 1$. The equivalence is not a discovery about LCSH; it is a consequence of using the dictionary sense boundary as the organizing unit for the heading lookup.

Will S generally match Y once this is avoided? Not necessarily, and there is principled reason to expect systematic, ambiguity-type-dependent divergence once heading assignment is done *blind* to the dictionary sense label — i.e., by mapping each sentence independently to the LCSH heading that best subject-catalogs its content, using only the sentence text and LCSH scope notes, without reference to which of Trott and Bergen's senses generated it. LCSH's granularity is governed by literary warrant and disciplinary/topical scope, not by the finer contrasts a psycholinguistic sense inventory draws (Broughton, 2012). Two predictions follow. First, homonymous contrasts (e.g., financial vs. river *bank*) typically correspond to entirely different subject domains, which LCSH's discipline-based structure is well suited to separate — blind coding should largely preserve $S = Y$ for homonyms. Second, polysemous contrasts (e.g., *lamb* the animal vs. the meat) are frequently too fine-grained for LCSH to register as distinct authority records at all; a single heading may legitimately cover both senses under blind coding, producing $S_i = 1$ where $Y_i = 0$. If this prediction holds, the S–Y correspondence itself becomes an empirically interesting, ambiguity-type-moderated quantity — directly continuous with H5 — rather than a tautology to be engineered away.

Three mitigations are available, operating at different levels of the design:

1. **Procedural (recommended for Experiment 1).** Replace sense-anchored heading lookup with blind, sentence-level LCSH assignment, ideally with an independent second coder and a documented reliability check (paralleling the two-librarian validation step considered for a parenthetical-qualifier extension). This directly targets the mechanism identified above and is a data-collection protocol change, not a change to the theoretical apparatus.
2. **Structural augmentation (already road-tested here).** Even where blind coding still yields high S–Y agreement for some words, hierarchical distance is a graded, continuously varying property of the LCSH BT/NT graph that is not fixed by construction the way a pair-relative same/different recoding is. Section 5.4.3 shows this variable carries genuine, non-tautological signal in exactly this dataset, which is why it is the more informative of the two structural predictors examined here regardless of how the S–Y question resolves under blind coding.
3. **Reconceptualizing the metric.** If sense-anchored mapping cannot be fully avoided for practical reasons (e.g., an existing corpus of pre-assigned headings), the categorical predictor should be redefined away from a pair-relative binary recoding and toward the heading identity itself as an independent multi-level label — computing $I(Y; H_{\text{prime}}, H_{\text{target}} \mid d)$ rather than $I(Y; S \mid d)$ for a derived indicator S that is definitionally downstream of Y. This does not repair a sense-anchored mapping, but it changes what the categorical warrant construct is measured against, and is worth stating explicitly as an open methodological

question for the full study rather than assuming the current S operationalization transfers unchanged.

5.4.3. The Hierarchical-Distance Result: A Non-Tautological Signal

Unlike S, h_i^{finite} (whether prime and target share any BT/NT ancestor within the search horizon) is not fixed by Y_i: of the 180 pairs with $h^{\text{finite}} = \text{TRUE}$, 55.6% (100 of 180) are same-sense and 44.4% (80 of 180) are different-sense; of the 119 pairs with $h^{\text{finite}} = \text{FALSE}$, none are same-sense (0 of 119). This is why the model could still be fit — with the caveats above — and why its result is the substantively interesting one from this proof of concept: $\overline{\text{CI}}_h = 0.190$ bits, $\bar{\eta}_h = 0.266$ (see Figure 5.2).[9] Hierarchical connectivity resolves just over a quarter of the disambiguation load left over after distance, without having been constructed to guarantee any particular answer.

Table 5.3: Piecewise cognitive information, h^{finite}-based full model. $n = 299$ pairs.

Bin	n_k	$\widehat{\text{CI}}_k$ (bits)	$P(d \in k)$	Contribution
$[0.028, 0.136]$	72	0.004	0.241	0.001
$(0.136, 0.243]$	74	0.367	0.247	0.091
$(0.243, 0.350]$	79	0.273	0.264	0.072
$(0.350, 0.458]$	32	0.151	0.107	0.016
$(0.458, 0.565]$	26	0.088	0.087	0.008
$(0.565, 0.673]$	16	0.046	0.054	0.002

[9] These are close to the pre-correction figures (0.197 bits, 0.273) despite the correction removing a large negative region from roughly the upper half of the distance range (Section 5.3.5) – the removed negative drag in bins 4–6 turns out to be offset almost exactly by a large *drop* in bins 1–2's contribution (bin 1: 0.096 $\to$ 0.004 bits), so the two effects largely cancel in the weighted sum. What changed substantively is the *shape*: the corrected per-bin estimates (0.004, 0.367, 0.273, 0.151, 0.088, 0.046 bits across the six bins) decline near-monotonically after an early peak, closely matching the theoretical inverted-U, whereas the pre-correction estimates were positive in bins 1–3 and negative in bins 4–6 – a categorically different, and non-theoretically-motivated, shape.

Table 5.3: Piecewise cognitive information, h^{finite}-based full model. $n = 299$ pairs.

Bin	n_k	$\widehat{\mathrm{CI}}_k$ (bits)	$P(d \in k)$	Contribution
			$\overline{\mathrm{CI}}_h$	**0.190 bits**
			$\bar{\eta}_h$	**0.266**

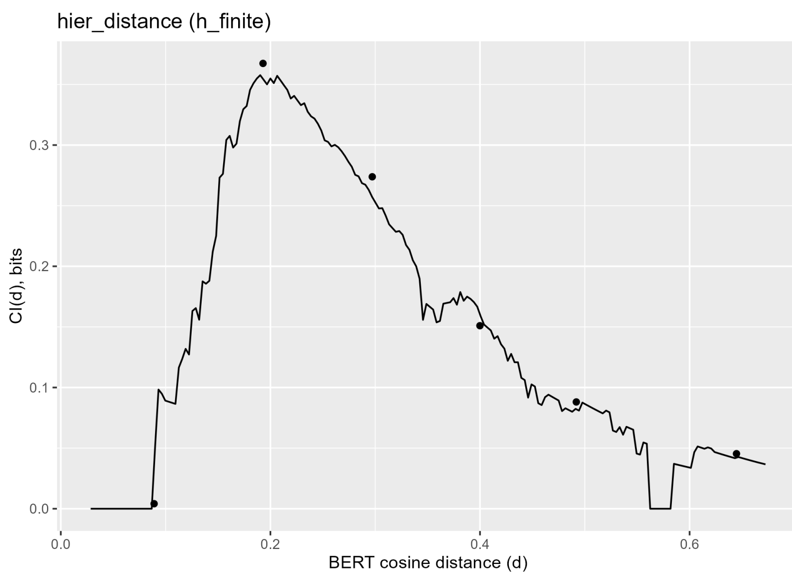


Figure 5.2: Cognitive information $\mathrm{CI}(d)$ as a function of BERT cosine distance d for the h^{finite} (`hier_distance`) full model (Firth-fit logistic; solid curve), computed via the same corrected, Jensen's-inequality-guaranteed decomposition as Figure 5.1 (Eq. 5.5), substituting h^{finite} for S. Points mark the piecewise binned estimates $\widehat{\mathrm{CI}}_k$ from Table 5.3 at each bin's mean distance. $\mathrm{CI}(d)$ peaks at ≈ 0.358 bits near $d \approx 0.190$ — closely matching the piecewise table's own bin-2 estimate (0.367 bits at $d = 0.193$), a stronger smooth/piecewise correspondence than the S-based model shows (Figure 5.1) — and declines near-monotonically thereafter, approaching but not reaching zero by the highest-distance bin. Unlike S (Figure 5.1), h^{finite} is not tautologically fixed by Y (Section 5.4.3), so this magnitude is a substantively informative estimate of LCSH hierarchical connectivity's disambiguation value, not a mechanical ceiling.

The pointwise diagnostic sharpens this further, though not in as clean a form as the naive check suggested. Under the corrected reference, 98 of the 299 pairs show negative pCI_i^h (minimum -0.911) — more than the 80 pairs flagged under the uncorrected pCI_i^h, and the relationship to $h^{\text{finite}} = \text{TRUE}, Y_i = 0$ pairs is no longer exact. Seventy of the original 80 remain negative; the other 10 — every *lamb* and *dinner* cross-sense pair ($h = 1$, the smallest possible nonzero hierarchical distance in this stimulus set) — are no longer negative, indicating their earlier flagging reflected the independently-fit baseline model's local misspecification rather than a genuine instance of the heading assignment misleading the model. The 28 newly negative pairs are, without exception, $h^{\text{finite}} = \text{TRUE}, Y_i = 1$ pairs — same-heading pairs, since $S_i \equiv Y_i$ in this dataset entails $h = 0$ whenever $Y_i = 1$ — with small magnitudes (-0.07 to -0.10 bits) concentrated at the low-distance edge of the lowest bin ($d \approx 0.12$–0.13 within the $[0.028, 0.136]$ bin). These are the model's most confidently *correct* pairs; their mildly negative pointwise value reflects the six-bin discretization itself, not a substantive disagreement between the heading assignment and the observed outcome, since a pair sitting near a bin's low edge inherits a marginal reference calibrated to the bin's blended average rather than its own specific distance.

Restricting to the 70 pairs whose negative status survives the correction, the ranking sharpens considerably: the most negative case is now *pick* ($h = 8$, $\mathrm{pCI}_h = -0.911$), followed by *charm* ($h = 7$, -0.838) and *lap* ($h = 10$, -0.797); two *hail* pairs[10] ($h = 6$) follow at -0.691 and -0.673. Notably, *port* ($h = 11$) — the single most negative case under the uncorrected pCI_h — no longer appears among the ten most negative pairs at all, for the same reason the *lamb*/*dinner* pairs dropped out: its earlier flagging was an artifact of the baseline reference, not a genuine anomaly relative to the true h^{finite}-marginalized rate. *pick*, *charm*, and *lap* are connected only through a common ancestor several hops away in the LCSH graph — the same qualitative

[10]The most negative cases (including *pick*, *lap*, and *hail*) are disproportionally verbs. Because LCSH headings are predominantly nominal in form, stimulus word senses that are verbs are systematically harder to map to subject headings than those that are nouns.

pattern the pre-correction diagnostic identified — but the specific items it implicates have changed, and the correction has also separated genuine misleading-assignment cases from a small cluster of low-distance-bin artifacts that the uncorrected version could not distinguish from them. This is a direct illustration of why Section 5.3.6's correction matters for the heading-reform diagnostic's practical use, not only for the aggregate consistency check.

5.4.4. Implications for the Plausibility of Cognitive Warrant

None of the results above bear on cognitive warrant directly: Y_i is Trott and Bergen's lexicographic sense classification, not a measurement of human behavior, and this chapter's pipeline never asks whether a person's processing is affected by an LCSH boundary. What the results do establish is a set of preconditions without which the planned behavioral test (Experiment 1) would not be worth running.

First, the baseline model leaves substantial residual uncertainty about sense identity at every distance short of the extremes (mean $H(Y \mid d) = 0.724$ bits across pairs) — the redundancy bound $\mathrm{CI}(d) \leq H(S \mid d)$ is nowhere close to binding in this dataset, meaning there is genuine informational room for a categorical variable to matter.

Second, when a structural LCSH variable is constructed so that it is not simply a re-encoding of the outcome, it carries non-trivial signal: h^{finite} recovers just over a quarter of the residual disambiguation load, a large and statistically well-supported effect ($\Delta\mathrm{Dev} = 89.43$, $p < .001$) by any conventional standard.

Third, the pipeline itself — logistic regression into binary entropy, piecewise and pointwise cognitive information, the redundancy bound, the heading-reform diagnostic — behaves exactly as specified on real linguistic material at full stimulus-set scale: it produces an interpretable, non-degenerate result when the input variable is informative, and it produces the theoretically correct degenerate ceiling, with all the standard statistical warning signs (separation, exploding SEs, deviance collapse), when the input variable is definitionally redundant with the outcome. A pipeline that could not distinguish these two cases would be a poor foundation for the full study; this one visibly can.

The S–Y finding is, if anything, the more instructive result for planning purposes, because it demonstrates a specific, previously implicit failure mode — sense-anchored heading mapping manufacturing agreement by construction — that a careless implementation of Experiment 1's LCSH-mapping protocol could reproduce silently. Surfacing it here, before any human data are collected, is exactly the kind of methodological return a computational proof of concept is meant to provide.

5.4.5. Implications for Moving Forward with Experiment 1

Taken together, these results support proceeding to the fully powered behavioral study, with three concrete design commitments that follow directly from this chapter's diagnostics rather than from the original Methods chapter alone.

First, LCSH heading assignment for Experiment 1's stimuli should be performed blind to any pre-existing sense classification, at the sentence level, with an independent reliability check, to avoid reproducing the $S \equiv Y$ artifact identified in Section 5.4.2; the ambiguity-type-moderated divergence this predicts (higher agreement for homonyms than polysemes) is itself worth recording as an exploratory measure.

Second, hierarchical distance should be retained as a primary structural predictor alongside, not merely as a supplement to, heading congruence — it is less susceptible to the construction artifact and, in this file, is the source of the only unambiguously non-tautological result — and it should be operationalized with a threshold or continuous specification rather than the coarse finite/infinite split used here.

Third, stimulus sampling for Experiment 1 should ensure denser coverage of the low end of the BERT distance range than Trott and Bergen's materials happen to provide: only by observing pairs close enough to $d = 0$ that the baseline model actually approaches certainty can the ascending half of the H3 inverted-U — and thus the full theoretical prediction, not only its descending half — be tested.

CHAPTER 6: CONCLUSION

6.1. What This Study Is, and What It Is Not

This study began with a question that library and information science has not had the methodology to answer: do the categorical boundaries that professional catalogers draw correspond to the boundaries inherent to human semantic processing? That question has an obvious home in LIS, where it bears on every claim about the utility of controlled vocabulary. But it also has a home in psycholinguistics—specifically, in the research program that Trott and Bergen (2023) inaugurated with their demonstration that word meaning is simultaneously categorical and continuous.

The connection is not incidental. Trott and Bergen showed that dictionary-derived boundaries predict processing difficulty above and beyond continuous distributional similarity. What they could not test—because their instrument (dictionary headword entries) did not support it—is whether categorical boundary effects depend on the *source* of the boundary. Dictionary sense boundaries are lexicographic constructs: they reflect the consensual judgment of lexicographers about when two meanings are distinct enough to warrant separate entries. LCSH boundaries are a different kind of construct: they reflect the professional judgment of catalogers about when two document types are distinct enough to warrant separate subject access points. The question of whether human processing responds to both in the same way is, at root, a question about the generalizability of hybrid meaning theory.

This study is a systematic test of that generalizability.

6.2. What the Framework Already Contributes to Hybrid Meaning Theory

Even before data collection, the cognitive warrant framework extends Trott and Bergen's theoretical program in three ways that are independent of the study's empirical outcomes.

The first is a formalization of the measurement problem. Trott and Bergen demonstrate that categorical and continuous components predict processing; they do not provide a framework for quantifying the *relative* contribution of categorical structure at each point in semantic space. $\mathrm{CI}(d) \equiv I(Y; S \mid d)$—the conditional mutual information between sense identity and category status given continuous distance—is precisely that framework. It produces a graduated scale for the categorical contribution of any boundary system, normalized to $[0, 1]$ via warrant efficiency $\eta(d) = \mathrm{CI}(d)/H(Y \mid d)$. This connects directly to the graduation problem that Trott & Parkinson-Coombs (2026) identify in the context of LLM benchmarking: $\mathrm{CI}(d)$ is a theoretically-derived functional mapping from a measured quantity to an underlying cognitive construct, with a principled zero and a principled maximum derived from information-theoretic first principles rather than assumed by convention.

The second is a mathematical account of the interaction structure. The inverted-U prediction (H3) is not a new empirical conjecture: it is a theorem derivable from $\mathrm{CI}(d)$'s formal properties under a single monotonicity assumption. This means that the linear interaction Trott and Bergen tested—and found null—is the wrong functional form for detecting the boundary's cognitive signature, not evidence that the signature is absent. If LCSH boundaries carry any categorical information at all (H2), the interaction *must* be nonlinear; the regression is asking where the peak falls and whether it is large enough to detect, not whether it exists.

The third is a structural account of the ambiguity-type null result. Trott and Bergen found no behavioral difference between homonyms and polysemes. Rather than treating this as a settled negative, the cognitive warrant framework offers an explanation: dictionary headword entries do not encode hierarchical distance, so there is no mechanism by which ambiguity type should modulate boundary effects in their design, and none was found. H6's mediation test—whether hierarchical distance absorbs the ambiguity type moderation once it is introduced as a predictor—is a direct test of this structural account. A confirmed H6 would retrospectively explain their null result without challenging the power or rigor of their study; it would instead identify the structural variable that their instrument could not provide.

6.3. What Completing the Study Would Add

The proof-of-concept analyses establish that the methodology is feasible: the LCSH mapping infrastructure produces valid stimulus materials, the hierarchical graph encodes meaningful structural variation, and the behavioral paradigms generate signal on the primary hypotheses.

What the full confirmatory study adds is inferential force—and a set of theoretical outcomes that would advance hybrid meaning theory in ways that cannot be achieved at current sample sizes.

A confirmed H2 at full power would establish that hybrid meaning processing generalizes to professionally imposed categorical boundaries: that the cognitive system responds to LCSH heading assignments as real sense partitions, not merely as a library artifact. This is a non-trivial generalization. Trott and Bergen's boundaries emerged from lexicographic consensus—a process that, whatever its limitations, is explicitly aimed at capturing psychologically real sense distinctions. LCSH boundaries emerge from literary warrant, cataloging rules, and institutional practice. If both produce the same categorical component in processing, this implies that the cognitive system is sensitive to categorical structure as such, not merely to the particular process that generated it.

A confirmed H3 at full power would add the first direct behavioral evidence for the boundary-distance interaction structure that the CI(d) framework entails. It would also provide an empirical estimate of where the peak of categorical sensitivity falls in distributional space for this stimulus domain—information that is theoretically useful for any subsequent study using BERT distance as a continuous predictor, because it identifies the region where categorical effects are most and least likely to be detectable.

A confirmed H6, combined with a confirmed E2-H5, would provide the strongest available evidence for the structural account of ambiguity-type effects: the same mediator absorbing the same apparent moderation across two independent behavioral paradigms. This outcome would close the ambiguity type question that Trott and Bergen's study left open—not by finding the effect they missed, but by showing why a well-powered study should not have found it.

Cross-experiment convergence (H7) would complete the construct validation argument. Item-level correlation between implicit disruption indices and explicit discrimination scores—across entirely different task demands, response modalities, and dependent variable structures—would establish cognitive warrant as a stable construct rather than a paradigm-specific measurement artifact. This is the multi-method triangulation that any measurement program needs before its instrument can be exported to new domains.

6.4. The Larger Picture

The mutual benefit that this research facilitates is asymmetric in the best possible way: the contributions to each field are qualitatively different, and neither comes at the expense of the other.

For psycholinguistics, this study offers a natural experiment that is structurally unavailable using standard lexicographic materials. LCSH's polyhierarchical DAG provides a continuous variable—hierarchical distance—that captures categorical proximity without collapsing it to a binary. Sentence boundaries in the design are defined by an institution rather than by lexicographic consensus, which decouples categorical structure from the semantic intuitions that necessarily influence dictionary construction. And the two-experiment design, with implicit and explicit DVs operationalizing the same latent construct, provides the kind of convergent validity evidence that process models of lexical access need to take categorical effects seriously as a general property of the architecture rather than an artifact of the task.

For library and information science, this study offers the first methodology for testing KOS design decisions against behavioral evidence. Whether LCSH's boundaries pass or fail the cognitive warrant test, the measurement instrument is the contribution—transferable to MeSH, FAST, the Art and Architecture Thesaurus, and any controlled vocabulary with recoverable hierarchical structure. The framework transforms KOS evaluation from a matter of professional consensus into a matter of empirical inquiry.

These two contributions share the same infrastructure. The LCSH mapping pipeline, the hierarchical graph, the behavioral paradigms, and the statistical framework were built to answer a question in LIS. They are, simultaneously, a psycholinguistics experiment on the generalizability of hybrid meaning theory. The data that would confirm or disconfirm cognitive warrant for LCSH are the same data that would extend or constrain hybrid meaning theory beyond the lexicographic boundary conditions under which it was originally established.

That convergence is not an accident of interdisciplinary opportunism. It follows from the structure of the problem: meaning is simultaneously a cognitive and an informational phenomenon, and the systems humans build to organize it are simultaneously products of professional practice and implicit theories about how cognition works. Testing those theories rigorously requires both the psycholinguistic paradigm and the knowledge organization infrastructure. This study brings them together for the first time.

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